

COMPENSATED GRAM CORRECTIONS FOR ZERO COUNTS OF THE RIEMANN ZETA FUNCTION

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ABSTRACT. We develop a compensated Gram-matrix correction for counting zeros of the Riemann zeta function. A single averaged matrix witness combines overlapping local windows without an outer-block loss. Its admissible local floor depends on the spectral clipping threshold; this dependence permits a stronger correction for the union of simple and critical-line zeros and for low-multiplicity counts than for the simple-critical intersection. Combining the finite argument with unconditional pair correlation and two computer-assisted seven-point inequalities gives lower limiting proportions exceeding 0.888554068993217696 for simple-or-critical zero mass, 0.837368508521536857 for mass of multiplicity at most two, and 0.939197203148497742 for locations of multiplicity at most two among distinct locations. Each improves an explicitly identified same-statistic comparison. We also obtain a simple-critical bound, a total-distinct bound, fixed higher-multiplicity-cutoff families, and qualified short-height extensions. The analytic deductions and the finite certification rules are given explicitly. Supplement information treats the numerical certificates and a local thirteen-point reduction; that reduction is not used as a whole-domain inequality or as a further zero-count improvement.

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1. INTRODUCTION

The distinction between a zero and a zero location is essential in questions about multiplicity. So is the distinction between being simple *and* lying on the critical line and being simple *or* lying there. These questions share analytic information, but they spend that information in different ways. A method that improves one count need not improve another. This work tries to make that distinction effective through a finite Gram-matrix argument.

The starting point is the unconditional pair-correlation theorem of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [4]. The recent arguments of Alpöge–Furman and Lamzouri [1, 11] connect this analytic input to the simultaneous simplicity and criticality of more than two thirds of the zeros. Local Gram corrections, including computer-assisted weighted overlap inequalities, subsequently appear in the public work of Ainta and Shi [2, 17]. Waddle’s public low-multiplicity draft [20] provides a further useful comparison: its scalar multiplicity accounting can be strengthened by retaining the entire simple-real Gram block.

Our contribution is a finite transfer mechanism and its application to these different counts. The main point is not to add unrelated local estimates to a spectral bound. Instead, all overlapping windows are represented inside one admissible matrix witness. The quadratic cost of that witness pays for their overlap. A row-sum estimate then controls the windows that require clipping. Since that control grows as the square of the clipping threshold, a local floor that is too large for the simple-critical argument can still be used for the union and multiplicity arguments.

There are two independent numerical inputs. One is Ainta’s seven-point cosine-kernel inequality, which we credit as prior work. The other uses an explicit two-mode perturbation of the same density. The perturbation raises the analytic energy, so its local improvement is useful only after both effects have been included in the same calculation. We give the exact formulas, rather than combining the energy of one kernel with the local certificate of another. The computer-assisted inputs are whole-domain inequalities, not sampled minima. Their finite acceptance rules and reproducibility requirements are supplied in the accompanying supplement.

All results in this article concern ordinary nontrivial zeros of ζ . Neither the Riemann hypothesis nor a narrow-box assumption on the zeros is used in the global conclusions. The short-height statements additionally invoke the stated theorem of Wang [21], currently a preprint; the global conclusions do not depend on it.

1.1. Counts and main conclusions. Write a nontrivial zero location as $\rho = \beta + i\gamma$, and write m_ρ for its multiplicity. All sums in this subsection run over distinct locations with $0 < \gamma \leq X$. Define

$$(1.1) \quad N(X) = \sum_{\rho} m_{\rho}, \quad n(X) = \#\{\rho : m_{\rho} = 1, \beta = \tfrac{1}{2}\},$$

$$(1.2) \quad N^d(X) = \sum_{\rho} 1, \quad U(X) = \sum_{m_{\rho}=1 \text{ or } \beta=1/2} m_{\rho},$$

$$(1.3) \quad M_{\leq r}(X) = \sum_{m_{\rho} \leq r} m_{\rho}, \quad D_{\leq r}(X) = \sum_{m_{\rho} \leq r} 1.$$

Thus U is a mass count of a union, whereas n counts the intersection. The denominator for $D_{\leq r}$ below is N^d , not N .

Here are the exact constants used throughout. Put

$$(1.4) \quad z = \frac{1}{\sqrt{2}}, \quad R_0 = \frac{1}{2} + z \cot z, \quad u_1 = \frac{13}{500}, \quad u_2 = -\frac{7}{500},$$

$$(1.5) \quad \Delta = \sum_{j=1}^2 \left(\frac{1}{2} - \frac{1}{4\pi^2 j^2} \right) u_j^2, \quad R_1 = R_0 + \Delta,$$

$$(1.6) \quad a_0 = \frac{19}{5000}, \quad d_0 = \frac{1}{500}, \quad a_1 = \frac{7}{400}, \quad d_1 = \frac{3}{250}, \quad T_1 = R_1 + d_1,$$

$$(1.7) \quad \Gamma = 4 + 2\sqrt{2(1-a_1)}, \quad Z = \frac{\Gamma^2}{8} = 3 - a_1 + 2\sqrt{2(1-a_1)}, \quad \beta = T_1 - 1 - a_1, \quad E = \frac{\beta}{Z}.$$

In particular $0 < E < 1/2$. The conclusions proved below are

$$(1.8) \quad \liminf_{X \rightarrow \infty} \frac{n(X)}{N(X)} \geq C := \frac{2 - R_0 - d_0}{1 - a_0} > 0.673058325315610967,$$

$$(1.9) \quad \liminf_{X \rightarrow \infty} \frac{N^d(X)}{N(X)} \geq \frac{1 + C}{2} > 0.836529162657805483,$$

$$(1.10) \quad \liminf_{X \rightarrow \infty} \frac{U(X)}{N(X)} \geq 1 - 2E > 0.888554068993217696,$$

$$(1.11) \quad \liminf_{X \rightarrow \infty} \frac{M_{\leq 2}(X)}{N(X)} \geq \frac{3 - T_1}{2 - a_1} > 0.837368508521536857,$$

$$(1.12) \quad \liminf_{X \rightarrow \infty} \frac{D_{\leq 2}(X)}{N^d(X)} \geq 1 - \frac{T_1 - 1 - a_1}{8 - 2T_1 - a_1} > 0.939197203148497742.$$

The strict decimal inequalities compare fixed real constants. They do not assert that every finite-height ratio already exceeds the printed decimal. For each fixed integer $r \geq 3$, the same argument also gives

$$(1.13) \quad \liminf_{X \rightarrow \infty} \frac{M_{\leq r}(X)}{N(X)} \geq 1 - \frac{r+1}{r-1}E,$$

$$(1.14) \quad \liminf_{X \rightarrow \infty} \frac{D_{\leq r}(X)}{N^d(X)} \geq 1 - \frac{E}{r-1-rE}.$$

The word “fixed” matters: no uniform asymptotic assertion with a height-dependent cutoff is intended.

1.2. Comparison with earlier bounds. The union, multiplicity-two mass, and multiplicity-two distinct-location conclusions improve identified unconditional bounds for the same counting statistics. The union bound improves Lamzouri’s Theorem 1.1 [11]; the two multiplicity-two bounds improve

the displayed formulas in Waddle’s public research draft [20]. The respective improvements, as differences of proportions, exceed

$$0.000934060819863356, \quad 0.001118156681831034, \quad 0.000469272388685036.$$

Multiplying these differences by 100 gives percentage-point gains. The comparisons use the exact expressions, not subtraction of rounded headline percentages.

To state the comparisons compactly, put

$$(1.15) \quad C_0 = 2 - R_0, \quad C_1 = \frac{3 - R_0}{2}, \quad B = (3 - 2\sqrt{2})(R_0 - 1), \quad C_2 = 1 - 2B.$$

Thus $C_0 = 0.672500703679411\dots$ is the Montgomery–Taylor simple-critical constant and $C_1 = 0.836250351839705\dots$ is the corresponding distinct-zero constant. Alpöge and Furman [1, Theorem A] established these bounds unconditionally by an inertia and rank–trace argument for a finite compression of Weil’s Hermitian form. Lamzouri [11] recovered them through a Hilbert-space argument and obtained the union constant

$$C_2 = \frac{1 + 2\sqrt{2} + 2C_0}{3 + 2\sqrt{2}} = 0.887620008173354\dots$$

Ratio	Present lower bound	Named comparison
U/N	$1 - 2E$ > 0.888554068993217696	Lamzouri [11]: $C_2 = 0.887620008173354\dots$
$M_{\leq 2}/N$	$\frac{3 - T_1}{2 - a_1}$ > 0.837368508521536857	Waddle [20]: $C_1 = 0.836250351839705\dots$
$D_{\leq 2}/N^d$	$1 - \frac{\beta}{8 - 2T_1 - a_1}$ > 0.939197203148497742	Waddle [20]: $1 - \frac{R_0 - 1}{2(4 - R_0)}$ $= 0.938727930759812\dots$
n/N	$\frac{2 - R_0 - d_0}{1 - a_0}$ > 0.673058325315610967	Devine [8]: $673399/10^6 = 0.673399$ (stated unconditional theorem)
N^d/N	$\frac{3 - R_0 - d_0 - a_0}{2(1 - a_0)}$ > 0.836529162657805483	Alpöge–Furman and Lamzouri [1, 11]: $C_1 = 0.836250351839705\dots$

TABLE 1. Five separate global lower limiting proportions. Each exact expression in the middle column bounds the corresponding lower limit from below; the strict inequalities compare these expressions with shorter decimals. All external statements listed here are unconditional. Waddle’s source is an unrefereed public draft and Devine’s is a public preprint. The last row is a comparison with the stated foundational bounds, not a claim of an overall distinct-zero record.

Here n counts simple critical zeros, whereas U counts with multiplicity the union of simplicity and critical-line location. The denominators N and N^d are different: $M_{\leq r}/N$ is a fraction of zero mass, and $D_{\leq r}/N^d$ is a fraction of distinct complex zero locations. Locations with the same imaginary part are not identified. These conventions also match the external statements in Table 1. A dyadic statement for additive counts gives the corresponding cumulative lower limit by summing sufficiently high dyadic pieces of $(0, X]$; the bounded initial remainder disappears after division by the same denominator.

For every fixed integer $r \geq 3$, Waddle’s displayed mass and distinct-location bounds are respectively

$$1 - \frac{(r+1)B}{r-1}, \quad 1 - \frac{B}{r-1-rB}.$$

Our corresponding formulas replace B by E , and the proved inequality $0 < E < B < 1/2$ makes both improvements strict. The mass gain is $(r+1)(B-E)/(r-1)$; the location comparison follows from the strict increase of $x/(r-1-rx)$ on the relevant interval. This conclusion uses the retained multiplicity budget. It is not a formal consequence of a union bound: a multiset supported on real locations of multiplicity $r+1$ has full union mass and no mass at multiplicity at most r .

The fixed-cutoff statement is an improvement over a specified family. It does not assert the best available bound for every cutoff. Simonič, Trudgian and Turnage-Butterbaugh [18, Theorems 4 and 7] give unconditional exponential estimates for the number of locations of exact multiplicity j . Weighting by j and summing the tail gives mass bounds tending to one as the cutoff grows, whereas the displayed mass family here tends to $1 - E < 1$. Consequently the exponential-tail bound is stronger for sufficiently large fixed cutoffs.

The simple-critical conclusion refines Ainta’s public bound [2], whose exact expression is

$$\frac{1345000C_0 - 2680}{1340003} = 0.673008527927779 \dots$$

It remains below Shi’s stated research-draft candidate $0.6733169771424713 \dots$ [17] and Devine’s stated 0.673399 [8]. The latter preprint’s numerical proof has not been independently reconstructed here. These are external comparison targets, not premises of our proof, and we claim no overall simple-critical or distinct-zero record. Zeta Lab’s public report [12] also records an eight-point value $0.67305298298962888 \dots$, distinguishing an interval-certified local inequality from its Lean implication conditional on that inequality. The public union extension [13] supplies another explicit treatment of the distinction between union and intersection.

Stronger numbers with additional hypotheses must be compared separately. The union bounds $8/9$ in [5, Remark 2] and $(2 + C_0)/3 = 0.890833567893 \dots$ discussed in [11, Remark 1.2] use a narrow-box assumption on the horizontal locations of all sufficiently high dyadic zeros; they are not unconditional union comparators. Under RH, Gonçalves, de Laat and Leijenhorst [9, Theorem 1] prove $\liminf M_{\leq 2}/N \geq 0.9614$. Their cutoff-three value 0.9787 also carries the theorem’s additional qualification concerning Maple summation of specified series. RH-dependent pair-correlation estimates in [7] include 0.6792 for simple zeros and 0.8477 for distinct zeros. Devine’s claims beyond 0.6792 likewise require additional density or horizontal-profile assumptions [8]. None of these numbers is an unconditional obstruction to the bounds proved here.

1.3. Relation to earlier methods. The analytic framework and several finite ingredients have established antecedents. Montgomery’s pair-correlation theorem [14] and the Montgomery–Taylor window reported in [15] underlie the constant R_0 ; the associated extremal problem is treated in [6]. Aryan [3] obtained an unconditional form of the relevant Fejér-kernel calculation, and Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [4] proved the published full-height pair-correlation formula used here. Lamzouri’s finite-multiset formulation and removal of the rational pair weight [11] provide the direct analytic interface. Wang [21] supplies the separate short-height input, with support strictly smaller than the fixed interval exponent. We prove no new pair-correlation theorem and do not insert a height-dependent test into a fixed-test error estimate.

Ainta [2] introduced the clipped spectral stability correction and quantitative overlapping seven-point windows used in this line of refinements. The Fourier perturbations of the cosine density and nonuniform pair weights with distance-capacity constraints in the inspected Ojha and Shi drafts [16, 17] are also prior work. Shi’s threshold-one trace-energy envelope, pressure rescaling, and two-certificate supporting-plane deduction are directly relevant antecedents. In particular, neither weighted overlap nor removal of the elementary threshold-one cap is claimed as a new idea here. The

arbitrary-parameter scalar multiplicity framework and its uncorrected low-multiplicity consequences are credited to the two-trace development and Waddle's displayed deductions [1, 20].

The finite contribution here is the use of one common averaged witness to transfer a whole-domain local Gram inequality directly to the clipped spectral defect while preserving its pressure floor. The uniform coefficient estimate depends on the clipping threshold. For the seven-point weights used here, its admissible cap is $40t^2/2401$. This dependence is material: the new local floor $a_1 = 7/400$ is unavailable through this cap at $t = 1$, but is admissible at $t = 2$ and at the union threshold $t = 1 + \sqrt{2(1 - a_1)}$. Thus the stronger local certificate is matched to the count being proved. Retaining the resulting simple-Gram correction in the separate per-location multiplicity costs then produces the mass and distinct-location bounds from that same certificate. The elementary variational identity for clipping is part of the proof; the gain comes from its common-witness application and the uniform capacity estimate.

The required two-mode profile, its normalization and energy cost, and the whole-domain seven-point inequality are established explicitly. The local interval engine is an attributed adaptation of Ainta's MIT-licensed implementation, using Arb ball arithmetic [10]. The separate cosine row for the simple-critical and total-distinct conclusions is Ainta's completed input. Our reproduced executions and exact scalar calculations certify the stated finite numerical premises; they do not replace the finite operator argument or the passage to zero limits.

Finally, bounded-potential versions of the common-witness argument give additional finite inequalities, but the general telescoping subaction idea already occurs in the public literature, including [8]. Partial larger-window certificates are identified by their domains. They supply no global zero-count conclusion unless the entire required configuration space has been covered.

1.4. Organization and the role of computation. We first prove the finite trace and inertia inequality, then construct the common local witness. A separate multiplicity calculation retains the simple Gram block while grouping all other locations by multiplicity. Only after those arguments are complete do we introduce the two kernels and invoke pair correlation. This order exposes the interface between finite linear algebra, local numerical certification and analytic number theory.

The numerical part has two logically separate jobs. Rigorous interval enclosures certify the local inequalities on the entire nonnegative gap orthant; exact rational interval arithmetic then encloses the constants in (1.8)–(1.14). The latter calculation does not replace the former. Conversely, a local certificate has no zero-count meaning until its matching kernel has passed through the analytic limit. The supplement explains the full acceptance mathematics, the finite coverage rules and the dependence on rigorous elementary-function arithmetic.

2. A FINITE TRACE INEQUALITY WITH A FREE CLIPPING THRESHOLD

2.1. The operator attached to a conjugation-invariant multiset. Let η be a real even square-integrable function of compact support, with $\int_{\mathbb{R}} \eta(s)^2 ds = 1$. Put $f = \eta^2$ and

$$(2.1) \quad K(z) = \int_{\mathbb{R}} f(s) e^{-2\pi i z s} ds, \quad v_z(s) = \eta(s) e^{-2\pi i z s}.$$

Compact support makes K entire. It is even, real on the real axis, and $K(0) = 1$. We take the inner product to be linear in its first variable, so that

$$\langle v_z, v_w \rangle = K(z - \bar{w}).$$

The rank-one notation $v \otimes w^*$ denotes the operator $h \mapsto \langle h, w \rangle v$.

Let $\mathcal{Z}$ be a finite conjugation-invariant multiset in $\mathbb{C}$, with total multiplicity N . A location and each of its repeated copies are distinguished when forming the following sum:

$$(2.2) \quad A = \sum_{z \in \mathcal{Z}} v_z \otimes v_z^*.$$

Conjugation invariance gives $A = A^*$, and every summand has trace one. All calculations may be made in the finite-dimensional span of the vectors that occur, without any assertion of their linear independence. In particular,

$$(2.3) \quad \text{Tr } A = N, \quad H := \|A\|_{\text{HS}}^2 = \sum_{z, w \in \mathcal{Z}} K(z - w)^2.$$

For the second identity, if $A_z = v_z \otimes v_z^*$, then $\text{Tr}(A_z A_w) = K(w - z)K(z - w) = K(z - w)^2$. Summation and selfadjointness give the identity. The square in this formula is a complex square, not the absolute square of each summand. Individual summands need not be nonnegative; the total is a squared Hilbert–Schmidt norm.

Let $x_1 \leq \dots \leq x_n$ be the simple real locations in $\mathcal{Z}$. Define

$$(2.4) \quad P = \sum_{j=1}^n v_{x_j} \otimes v_{x_j}^*, \quad M = (K(x_i - x_j))_{1 \leq i, j \leq n}, \quad Q = A - P.$$

The Gram matrix M is positive semidefinite and has diagonal one. Its order is n , even if its rank is smaller. The nonzero spectra of M and P agree, and $\text{Tr } P = n$.

2.2. Exceptional orbits and positive inertia. Let r_R count the multiple real locations, of respective multiplicities $k_j \geq 2$, and let r_C count the unordered nonreal conjugate pairs, of respective multiplicities $\ell_j \geq 1$ at each member of a pair. Set $b = r_R + r_C$. Each real multiple contributes one positive rank-one term to Q . For a nonreal pair write

$$g_z = (v_z + v_{\bar{z}})/2, \quad h_z = (v_z - v_{\bar{z}})/(2i).$$

Its contribution is

$$\ell_z(v_z \otimes v_z^* + v_{\bar{z}} \otimes v_{\bar{z}}^*) = 2\ell_z(g_z \otimes g_z^* - h_z \otimes h_z^*).$$

Consequently $Q = Q_a - Q_b$ with $Q_a, Q_b \geq 0$ and $\text{rank } Q_a \leq b$. On $\ker Q_a$ the quadratic form of Q is nonpositive, so the number of its positive eigenvalues satisfies

$$(2.5) \quad n_+(Q) \leq b.$$

These nonorthogonal orbit terms are used only to prove this inertia bound. The square completion below instead uses the spectral positive and negative parts of Q , which are orthogonal.

For $t \geq 0$ define the clipped Gram defect

$$(2.6) \quad D_t(M) = \text{Tr}((M - I)^2 - (M - (1 + t)I)_+^2).$$

Here the positive part is defined spectrally. Expanding the first square and retaining the zero eigenvalues of M gives the exact identity

$$(2.7) \quad D_t(M) = \text{Tr } P^2 - n - \text{Tr}(P - (1 + t)I)_+^2.$$

The value is nonnegative: for an eigenvalue $x \geq 0$ of M , the scalar function $(x - 1)^2 - (x - 1 - t)_+^2$ is nonnegative.

For every $\Gamma \geq 2$, with $t = \Gamma/2 - 1$, we now obtain

$$(2.8) \quad H \geq \Gamma N - (\Gamma - 1)n - \frac{\Gamma^2}{4}b + D_t(M).$$

Indeed, write $Q = Q_+ - Q_-$ for its spectral decomposition. Subtracting $\Gamma \text{Tr } A$ from $\text{Tr } A^2$ gives

$$H - \Gamma N = \text{Tr } P^2 - \Gamma n + \text{Tr}(Q_+^2 - \Gamma Q_+) + \text{Tr}(Q_-^2 + \Gamma Q_- - 2PQ_-) + 2 \text{Tr}(PQ_+).$$

The last term is nonnegative. The first quadratic expression in Q_+ is at least $-\Gamma^2 b/4$, by (2.5) and scalar square completion. For a selfadjoint X and $B \geq 0$,

$$(2.9) \quad \text{Tr}(B^2 - 2XB) \geq -\text{Tr } X_+^2.$$

To see this, diagonalize X : off-diagonal entries of B add a nonnegative amount to $\text{Tr } B^2$, and each nonnegative diagonal entry minimizes its scalar quadratic at the positive part of the corresponding eigenvalue of X . Apply this with $X = P - \Gamma I/2$. Equation (2.7) completes the proof of (2.8).

2.3. The excess that the different counts measure. Write

$$(2.10) \quad u = N - n, \quad e = u - 2b = \sum_{j=1}^{r_R} (k_j - 2) + 2 \sum_{j=1}^{r_C} (\ell_j - 1) \geq 0.$$

Use the definitions of $U, N^d, M_{\leq r}, D_{\leq r}$ from the introduction, with “critical” replaced for now by “real.” The mass excluded from the union consists precisely of the nonreal pairs with $\ell_j \geq 2$. Therefore

$$(2.11) \quad N - U = 2 \sum_{\ell_j \geq 2} \ell_j \leq 2e.$$

For $r \geq 2$, writing $L_{>r} = N - M_{\leq r}$, we similarly have

$$(2.12) \quad L_{>r} \leq \frac{r+1}{r-1} e.$$

For a real location of multiplicity $k \geq r+1$, this follows from $k/(k-2) \leq (r+1)/(r-1)$. A nonreal pair of multiplicity $\ell \geq r+1$ contributes 2ℓ to the mass and $2(\ell-1)$ to e , with an even smaller ratio. Summing proves the assertion.

The distinct count has a separate exact identity:

$$(2.13) \quad N - N^d = e + r_R, \quad N^d \geq \frac{N + n - e}{2}.$$

The second assertion uses $r_R \leq b = (N - n - e)/2$. In particular it is not an inference from a bound for n alone: the excess term must be retained until the trace inequality is substituted.

Suppose, for some $a, d, B \geq 0$, a local argument has established

$$(2.14) \quad D_t(M) \geq an - dS - B, \quad S = x_n - x_1 \quad (n \geq 2), \quad S = 0 \quad (n \leq 1).$$

Substituting $b = (u - e)/2$ into (2.8) yields

$$(2.15) \quad H \geq (1+a)N - dS - B + \left(\Gamma - 1 - \frac{\Gamma^2}{8} - a \right) u + \frac{\Gamma^2}{8} e.$$

At $\Gamma = 4$ this is equivalently

$$(2.16) \quad (1-a)n \geq 2N - H - dS - B + 2e.$$

For the union, it is useful instead to choose

$$(2.17) \quad \Gamma = 4 + \sqrt{8(1-a)}, \quad 0 \leq a < 1,$$

which makes the coefficient of u exactly zero. This algebraic choice, and not a numerical sign test near zero, will be used in the limit.

3. ONE WITNESS FOR ALL OVERLAPPING WINDOWS

3.1. The variational formula. Let $G \geq 0$ be an $n \times n$ Hermitian matrix with diagonal one, and write $Y = G - I$. For $t \geq 1$,

$$(3.1) \quad D_t(G) = \max_{T=T^*, \|T\|_{\text{op}} \leq t} \{2 \text{Tr}(YT) - \text{Tr}(T^2)\}.$$

After diagonalizing Y , deleting the off-diagonal entries of T does not change the linear term, reduces its squared norm, and preserves the constraint on its diagonal. The scalar maximum is $y^2 - (y-t)_+^2$ because $y \geq -1 \geq -t$. This proves the formula. The restriction $t \geq 1$ is essential here: a symmetric norm ball would also clip some negative eigenvalues if $t < 1$.

Fix $m \geq 2$ and nonnegative symmetric coefficients c_{ij} , with zero diagonal, such that

$$(3.2) \quad \sum_{i=1}^{m-h} c_{i,i+h} \leq 2 \quad (1 \leq h < m), \quad C_{\max} := \max_i \sum_{j \neq i} c_{ij} > 0.$$

For each full consecutive block B of length m , use local indices $1, \dots, m$ and define

$$(3.3) \quad V_B = \sum_{i < j} c_{ij} |G_{B_i B_j}|^2, \quad (Z_B)_{ij} = \frac{m}{2} c_{ij} G_{B_i B_j} \quad (i \neq j), \quad (Z_B)_{ii} = 0,$$

$$(3.4) \quad s_B = \max_i \sum_j |(Z_B)_{ij}|, \quad \nu_B = \min(1, t/s_B), \quad T_B = \nu_B Z_B,$$

where $\nu_B = 1$ when $s_B = 0$. The row-sum bound gives $\|T_B\|_{\text{op}} \leq t$.

There are m partitions of the integer line into consecutive m -blocks, one for each residue of the first index modulo m . Restrict them to $\{1, \dots, n\}$, placing T_B on every full block and zero on each incomplete endpoint block. The resulting m block-diagonal matrices all have norm at most t . Their average $\bar{T}$ is therefore admissible in (3.1). Every full consecutive window occurs exactly once in this average.

The linear term is

$$2 \text{Tr}(Y \bar{T}) = 2 \sum_B \nu_B V_B.$$

For a fixed global pair of indices of separation h , its local coefficients among the windows containing it form a subcollection of $c_{1,1+h}, \dots, c_{m-h,m}$. Their sum is at most two. Weighted Cauchy–Schwarz consequently gives

$$\left(\sum_{B \ni i,j} c_{ij}^B \nu_B \right)^2 \leq 2 \sum_{B \ni i,j} c_{ij}^B \nu_B^2.$$

Since the corresponding entry of $\bar{T}$ is $\frac{1}{2} G_{ij} \sum_{B \ni i,j} c_{ij}^B \nu_B$, summing its absolute square over both orientations of each pair yields

$$\text{Tr} \bar{T}^2 \leq \sum_B \nu_B^2 V_B.$$

Thus one admissible witness proves the simultaneous estimate

$$(3.5) \quad D_t(G) \geq \sum_B J_t(B), \quad J_t(B) := (2\nu_B - \nu_B^2) V_B.$$

This is where the overlap is paid for. No sum of separate spectral pinchings, and no subsequent outer-block envelope, is required.

3.2. The threshold-dependent cap and pressure transfer. The row sum in (3.4) is controlled by the very energy that is being retained. For each row, weighted Cauchy–Schwarz gives

$$s_B^2 \leq \frac{m^2}{4} C_{\max} V_B.$$

Put

$$(3.6) \quad b_m = \frac{4}{m^2 C_{\max}}.$$

If $s_B \leq t$, then $J_t(B) = V_B$. If $s_B \geq t$, then

$$(3.7) \quad J_t(B) = \left(\frac{2t}{s_B} - \frac{t^2}{s_B^2} \right) V_B \geq b_m (2ts_B - t^2) \geq b_m t^2.$$

Now take $G_{ij} = K(x_i - x_j)$ for real ordered points, and let $g_j = x_{j+1} - x_j \geq 0$. Suppose the local kernel inequality

$$(3.8) \quad \sum_{i < j} c_{ij} |K(g_i + \cdots + g_{j-1})|^2 + \sum_{j=1}^{m-1} p_j g_j \geq \epsilon \quad (g_j \geq 0)$$

holds with $p_j \geq 0$ and

$$(3.9) \quad 0 \leq \epsilon \leq b_m t^2.$$

On an unclipped block, (3.8) bounds J_t directly; on a clipped block, (3.7) and nonnegative pressure do so. Sum over all $n - m + 1$ full windows when $n \geq m$. Each elementary gap is charged by at most one copy of each p_j , so its total coefficient is at most $P_* := \sum_{j=1}^{m-1} p_j$. Equation (3.5) then proves

$$(3.10) \quad D_t(G) \geq \epsilon n - P_* S - (m - 1)\epsilon.$$

For $n < m$, the right side is nonpositive and the assertion follows from $D_t \geq 0$. This also treats empty and one-point Gram matrices. The endpoint term is retained until the eventual height limit.

Only the unclipped domain $s_B \leq t$ needs (3.8); a whole-orthant certificate is a convenient stronger premise. The threshold condition nevertheless cannot be omitted. For the uniform choice

$$(3.11) \quad c_{i,i+h} = \frac{2}{m-h}, \quad C_{\max} = 2 \sum_{h=1}^{m-1} \frac{1}{h},$$

the endpoint row realizes the maximum. At $m = 7$,

$$(3.12) \quad b_7 = \frac{40}{2401}.$$

Our cosine floor a_0 is smaller than b_7 , whereas the two-mode floor a_1 is larger. The latter is therefore not transferred at $t = 1$ by this argument. It is transferred at $t = 2$, and at the union threshold $t = 1 + \sqrt{2(1 - a_1)} > 2$.

3.3. A bounded state correction. The common witness also permits a bounded correction that telescopes between neighboring windows. This observation is useful without asserting that an advantageous correction has been found for any particular unresolved numerical problem.

Put $q = m - 1$, and let $\Phi : \mathbb{R}_{\geq 0}^{q-1} \rightarrow \mathbb{R}$ have finite oscillation $\Omega = \sup \Phi - \inf \Phi$. For $q = 1$ its domain is a singleton. Keep $t \geq 1$, $p_j \geq 0$ and $\epsilon \geq 0$. Suppose that for every nonnegative q -tuple,

$$(3.13) \quad J_t(g) + \sum_{j=1}^q p_j g_j \geq \epsilon + \Phi(g_1, \dots, g_{q-1}) - \Phi(g_2, \dots, g_q).$$

The state differences telescope once across the complete sequence of windows. Consequently

$$(3.14) \quad D_t(G) \geq \epsilon n - P_* S - q\epsilon - \Omega.$$

For $n < m$ the right side is nonpositive and $D_t \geq 0$ again suffices. No regularity of Φ is needed. If the corresponding inequality with V in place of J_t is known only for $s \leq t$, the sufficient condition $\epsilon + \Omega \leq b_m t^2$ supplies the complementary domain using (3.7).

There is also a useful uniform stability bound. Suppose two normalized nonnegative densities have Fourier transforms differing by at most ρ on the real line. Both transforms have absolute value at most one, so (3.2) gives $|V - V'| \leq 4q\rho$ and $|s - s'| \leq mC_{\max}\rho/2$. The function $s \mapsto \min(1, t/s)$, continuously extended at zero, is $1/t$ -Lipschitz. Since $V' \leq 2q$, it follows that

$$(3.15) \quad |J_t - J'_t| \leq \left(4q + \frac{2qmC_{\max}}{t}\right) \rho.$$

Thus a proved joined inequality can be smoothed with a controlled floor loss. One must smooth that full inequality, not assume that the boundary $s = t$ remains fixed. A dilation by $\sigma > 0$ replaces Φ by $\Phi(\sigma \cdot)$ and leaves its oscillation unchanged.

4. MULTIPLICITY ACCOUNTING WITH THE SIMPLE GRAM BLOCK RETAINED

Let x_m count real locations of multiplicity m and let y_ℓ count unordered nonreal pairs of multiplicity ℓ at each member. Then $n = x_1$, $N = \sum_m m x_m + 2 \sum_\ell \ell y_\ell$, and $N^d = \sum_m x_m + 2 \sum_\ell y_\ell$. For $c \geq 1$ define

$$(4.1) \quad h_c(x) = c^2 - (c - x)_+^2 \quad (x \geq 0).$$

This is nondecreasing and concave, and $h_c(0) = 0$.

Include *all* real locations in the positive operator $P_R = \sum_{z \in \mathcal{Z} \cap \mathbb{R}} v_z \otimes v_z^*$, with multiplicity. The positive inertia of $A - P_R$ is at most $y := \sum_\ell y_\ell$. Repeating the square completion in Section 2, now with $\Gamma = 2c$, gives

$$(4.2) \quad H \geq 2cN - \text{Tr } h_c(P_R) - c^2 y.$$

For clarity, the contribution of an eigenvalue x of P_R is $x^2 - 2cx - (x - c)_+^2 = -h_c(x)$, which proves the displayed form.

Use one weighted Gram column $\sqrt{m} v_z$ per distinct real location of multiplicity m . Pinch this Gram matrix into the entire simple-real block M and one scalar block $[m]$ per multiple-real location. Scalar concavity gives

$$(4.3) \quad \text{Tr } h_c(P_R) \leq \text{Tr } h_c(M) + \sum_{m \geq 2} h_c(m) x_m.$$

Indeed, diagonalize each compressed block and use its eigenvectors as an orthonormal basis of the whole Gram space. Jensen's inequality $h_c(\langle Gv, v \rangle) \geq \langle h_c(G)v, v \rangle$ follows from the spectral measure of G at v . Summing proves (4.3). Operator concavity is not required. Gram zero eigenvalues cause no discrepancy because $h_c(0) = 0$.

Since $\text{Tr } M = n$, the exact simple-block identity is

$$\text{Tr } h_c(M) = (2c - 1)n - D_{c-1}(M).$$

Combining this with (4.2) and (4.3) produces the corrected multiplicity inequality

$$(4.4) \quad H \geq 2cN - \sum_{m \geq 1} h_c(m) x_m - c^2 \sum_{\ell \geq 1} y_\ell + D_{c-1}(M).$$

The scalar master without the final Gram term is the comparison framework in [20]. The retained term is indispensable for the improvements made here.

4.1. Mass of multiplicity at most two. Take $c = 3$, and suppose $D_2(M) \geq an - dS - B$ with $0 \leq a \leq 1/2$. In (4.4), the real-location debits are $5 - a$ for $m = 1$, 8 for $m = 2$, and 9 for $m \geq 3$; the debit per nonreal pair is 9 . Per unit mass of multiplicity at most two, these are at most $5 - a$: the four possible values are $5 - a, 4, 9/2, 9/4$. For all higher multiplicities the debit per unit mass is at most 3 . Hence

$$H + dS + B \geq 6N - (5 - a)M_{\leq 2} - 3(N - M_{\leq 2}),$$

and therefore

$$(4.5) \quad M_{\leq 2} \geq \frac{3N - H - dS - B}{2 - a}.$$

This is a mass estimate; it is not yet a proportion among distinct locations.

4.2. Costs per location and the distinct denominator. Assume a finite upper budget

$$(4.6) \quad H + dS + B \leq \mathcal{T}N, \quad \alpha = 2c - \mathcal{T}, \quad \beta = \mathcal{T} - 1 - a.$$

With $D_{c-1}(M) \geq an - dS - B$, equation (4.4) implies

$$(4.7) \quad \sum_{m \geq 1} \lambda_m x_m + \sum_{\ell \geq 1} \omega_\ell y_\ell \leq 0, \quad \lambda_m = \alpha m - h_c(m) + a \mathbf{1}_{m=1}, \quad \omega_\ell = 2\alpha\ell - c^2.$$

Suppose that for a fixed cutoff r and some $\gamma_r > 0$, these costs are at least $-\beta$ per low-multiplicity location and at least $\gamma_r - \beta$ per high-multiplicity location. A nonreal pair contains two locations, so its required bounds are twice the corresponding quantities. Then (4.7) gives $0 \geq -\beta N^d + \gamma_r(N^d - D_{\leq r})$, whence

$$(4.8) \quad \frac{D_{\leq r}}{N^d} \geq 1 - \frac{\beta}{\gamma_r}.$$

This deduction requires $N^d > 0$, as will hold at sufficiently large heights.

For $r = 2$ and $c = 3$, sufficient conditions are

$$(4.9) \quad \beta \geq 0, \quad \alpha = 6 - \mathcal{T} > 0, \quad 3 - \mathcal{T} - a \geq 0, \quad 1 - 2a \geq 0, \quad \gamma_2 = 8 - 2\mathcal{T} - a > 0.$$

Here $\lambda_1 = -\beta$, $\lambda_2 + \beta = 3 - \mathcal{T} - a$, and for $m \geq 3$

$$\lambda_m + \beta - \gamma_2 = \alpha(m - 3) \geq 0.$$

For pairs, $\omega_1 + 2\beta = 1 - 2a \geq 0$, and increasing ℓ only increases the cost. At the first high multiplicity, $\omega_3 + 2\beta - 2\gamma_2 = 9$, which remains positive for $\ell \geq 3$. These facts check every multiplicity, not a finite selection of them, and give

$$(4.10) \quad \frac{D_{\leq 2}}{N^d} \geq 1 - \frac{\mathcal{T} - 1 - a}{8 - 2\mathcal{T} - a}.$$

For $r \geq 3$, let $3 < c < 4$, and assume

$$(4.11) \quad \alpha > 0, \quad \beta \geq 0, \quad 3 - \mathcal{T} - a \geq 0, \quad 8 - 2\mathcal{T} - a \geq 0,$$

$$(4.12) \quad 4\alpha + \beta - c^2 \geq 0, \quad q_c := 2c - 1 - c^2/2 - a \geq 0, \quad \gamma_r := (r + 1)\alpha + \beta - c^2 > 0.$$

For $m = 1, 2, 3$, the values of $\lambda_m + \beta$ are respectively 0, $3 - \mathcal{T} - a$, and $8 - 2\mathcal{T} - a$. For $m \geq 4$, $\lambda_m + \beta = \alpha m + \beta - c^2$, which is nonnegative at $m = 4$ and increasing. At $m = r + 1$ it equals γ_r . For a nonreal pair the smallest value of $\omega_\ell + 2\beta$ is $2q_c \geq 0$. At $\ell = r + 1$ it exceeds $2\gamma_r$ by exactly c^2 . Thus (4.8) holds for every fixed $r \geq 3$ under the displayed conditions.

At $2c = \Gamma$ from (2.17), write $Z = \Gamma^2/8$ and $E = \beta/Z$. The identity $\Gamma - 1 - Z - a = 0$ gives $\alpha = Z - \beta$ and

$$(4.13) \quad \gamma_r = Z(r - 1 - rE).$$

This explains the denominator in (1.14). It is obtained by per-location accounting, not by dividing a mass estimate by a distinct count.

5. THE TWO KERNELS AND THEIR CERTIFIED LOCAL INEQUALITIES

5.1. Normalization and the exact energy cost. On $[-1/2, 1/2]$, define

$$(5.1) \quad f_0(s) = \frac{\cos(\sqrt{2}s)}{\sqrt{2}\sin z}, \quad f_1(s) = f_0(s) + u_1 \cos(2\pi s) + u_2 \cos(4\pi s), \quad z = 1/\sqrt{2},$$

and extend both functions by zero outside that interval. The coefficients u_1, u_2 are those in (1.4). Direct integration gives $\int f_0 = \int f_1 = 1$. These are nonnegative densities, not signed test functions: on the interval $\cos(\sqrt{2}s) \geq 3/4$ and $\sqrt{2}\sin z \leq 1$, so $f_0 \geq 3/4$ and $f_1 \geq 3/4 - |u_1| - |u_2| = 71/100 > 0$.

For a real even compactly supported density $f \in L^1 \cap L^2$, define

$$(5.2) \quad \mathcal{R}(f) = \int_{\mathbb{R}} f(s)^2 ds + \iint_{\mathbb{R}^2} |s - t| f(s) f(t) ds dt.$$

The energy of f_0 is R_0 in (1.4). A short calculation also explains why its perturbation cost is exactly quadratic. On the support interval put $W_0(s) = \int |s - t| f_0(t) dt$. Then $W_0'' = 2f_0$ and $f_0'' = -2f_0$. Evenness shows that $f_0 + W_0$ is constant. Its value at $s = 1/2$ is $z \cot z + 1/2 = R_0$, so

$$\mathcal{R}(f_0) = \int f_0(f_0 + W_0) = R_0.$$

If h has integral zero on the interval, the linear variation of $\mathcal{R}(f_0 + h)$ vanishes. Write $H_h(s) = \int_{-1/2}^s h(t) dt$, so H_h vanishes at both endpoints. With $W_h(s) = \int |s - t| h(t) dt$ one has $W_h' = 2H_h$ and hence, by integration by parts,

$$\iint |s - t| h(s) h(t) ds dt = -2 \int H_h(s)^2 ds.$$

For $h(s) = \sum_j u_j \cos(2\pi j s)$, trigonometric orthogonality therefore gives

$$(5.3) \quad \mathcal{R}(f_0 + h) - R_0 = \int h^2 - 2 \int H_h^2 = \sum_j \left(\frac{1}{2} - \frac{1}{4\pi^2 j^2} \right) u_j^2.$$

The functional is quadratic, so there is no omitted higher-order remainder. In particular $\mathcal{R}(f_1) = R_1 = R_0 + \Delta$.

The two pieces of the energy must be kept separately when the support is dilated. Put $A_j = \int f_j^2$ and $B_j = R_j - A_j$ for $j = 0, 1$. Direct integration yields

$$(5.4) \quad A_0 = \frac{1}{4 \sin^2 z} + \frac{z \cot z}{2},$$

$$(5.5) \quad A_1 = A_0 + 2 \sum_{j=1}^2 \frac{(-1)^{j+1} u_j}{2\pi^2 j^2 - 1} + \frac{1}{2} \sum_{j=1}^2 u_j^2.$$

For example, the cross term follows from $\int f_0(s) \cos(2\pi j s) ds = (-1)^{j+1} / (2\pi^2 j^2 - 1)$. If $f_{j,\sigma}(s) = \sigma^{-1} f_j(s/\sigma)$, then

$$(5.6) \quad \mathcal{R}(f_{j,\sigma}) = R_j(\sigma) := \frac{A_j}{\sigma} + B_j \sigma.$$

Both coefficients are positive. The use of A_1, B_1 , rather than the cosine coefficients A_0, B_0 , is necessary for the perturbed short-height calculation.

5.2. The local inputs. Let $k_j = \widehat{f_j}$ with the Fourier convention (2.1). Writing $\text{sinc}(x) = \sin(x)/x$ with value one at zero, their entire formulas are

$$(5.7) \quad k_0(x) = \frac{\text{sinc}(\pi x - z) + \text{sinc}(\pi x + z)}{2\sqrt{2} \sin z},$$

$$(5.8) \quad k_1(x) = k_0(x) + \frac{1}{2} \sum_{j=1}^2 u_j \{ \text{sinc}(\pi(x - j)) + \text{sinc}(\pi(x + j)) \}.$$

Define the seven-point energy for six nonnegative gaps by

$$(5.9) \quad V_{7,j}(g) = \sum_{h=1}^6 \frac{2}{7-h} \sum_{i=1}^{7-h} k_j(g_i + \dots + g_{i+h-1})^2.$$

The two computer-assisted local inequalities are

$$(5.10) \quad V_{7,0}(g) + \frac{1}{3000} \sum_{i=1}^6 g_i \geq \frac{19}{5000},$$

$$(5.11) \quad V_{7,1}(g) + \frac{1}{500} \sum_{i=1}^6 g_i \geq \frac{7}{400}.$$

Both hold for every $g \in \mathbb{R}_{\geq 0}^6$, including zero gaps and unbounded configurations. Equation (5.10) is Ainta's certificate [2]. Equation (5.11) is the two-mode certificate used for the present union and multiplicity bounds. The supplement gives the rigorous finite verification in detail.

Here is the structure of that verification, to clarify what is asserted. Nonnegative pressure first reduces a possible strict counterexample to a compact simplex. A single-coordinate lower test cuts this down further. Outward scalar enclosures on closed grid cells give lower bounds for every squared-kernel term. A sum of h gaps requires all h final scalar cells; the extra $h - 1$ indices at a summed upper endpoint are included. Adaptive binary subdivision covers the resulting boxes. Interval energy, whole-box Taylor bounds and rigorous positive-definiteness tests each supply sufficient acceptance rules. Removable singularities in (5.7)–(5.8) use convergent power series with explicit remainders. An unaccepted terminal box, a time limit or a failed arithmetic guard gives no certificate. The recorded completed covers exhaust their domains under these rules. This is stronger than checking the objective at grid points, and the proof does not treat a digest or a node count as a substitute for coverage.

In (3.10), these rows give $a = a_j$ and $d = 6p = d_j$, with endpoint debit $B = 6a_j$. The first may be used at $t = 1$. For the second,

$$\frac{a_1}{b_7} = \frac{16807}{16000} > 1, \quad a_1 < 4b_7 < b_7(1 + \sqrt{2(1 - a_1)})^2.$$

Thus its admissible thresholds include the two required in Sections 2 and 4. These are separate uses of its *same* local certificate, not sums of corrections from different densities.

6. FROM THE FINITE OPERATOR TO ZETA ZEROS

6.1. The fixed-test pair-correlation input. We state precisely the analytic theorem being used. If g is a fixed real even integrable function supported in $[-1, 1]$ and Lipschitz at zero, the test-function form of [4, Lemma 5], also stated in [11, Lemma 3.1], is, with $L = \log X$,

$$(6.1) \quad \sum_{0 < \gamma, \gamma' \leq X} \hat{g}\left(\frac{i(\rho - \rho')L}{2\pi}\right) \frac{4}{4 - (\rho - \rho')^2} = \frac{XL}{2\pi} \left\{ g(0) + \int_{\mathbb{R}} |\alpha| g(\alpha) d\alpha + O_g(L^{-1/2}) \right\}.$$

The zeros are counted with multiplicity, and the formula is unconditional. We use it only for fixed smooth functions supported strictly inside $(-1, 1)$. No new proof of (6.1) is claimed here.

Take a real even smooth η supported strictly in $(-\sigma/2, \sigma/2)$, with $0 < \sigma < 1$ and $\int \eta^2 = 1$. Set $f = \eta^2$, $K = \hat{f}$ and $G = f * f$. Both G and G'' are admissible fixed tests. For $w = i(\rho - \rho')L/(2\pi)$, integration by parts gives $\widehat{G''}(w) = -4\pi^2 w^2 K(w)^2$. Consequently the exact identity

$$(6.2) \quad K(w)^2 = \left(\widehat{G}(w) - \frac{\widehat{G''}(w)}{4L^2} \right) \frac{4}{4 - (\rho - \rho')^2}$$

removes the rational weight. Its denominator is nonzero: the real part of the difference of two nontrivial zeros lies in $(-1, 1)$, so that difference cannot be 2 or -2 .

Apply (6.1) *separately* to G and G'' , then combine their sums by (6.2). The second term is $O_f(X/L)$, while the first has leading coefficient

$$G(0) + \int |\alpha| G(\alpha) d\alpha = \mathcal{R}(f).$$

Evenness of f changes the double integral with $|s+t|$ into that with $|s-t|$. Thus, for the multiset

$$(6.3) \quad \mathcal{Z}_X = \left\{ \frac{i(\rho - 1/2) \log X}{2\pi} : 0 < \gamma \leq X \right\},$$

equation (2.3) satisfies

$$(6.4) \quad \frac{H_f(X)}{N(X)} \rightarrow \mathcal{R}(f).$$

The functional equation and conjugation of ζ preserve the multiplicity under $\rho \mapsto 1 - \bar{\rho}$. That map acts as conjugation on (6.3); its real elements are exactly the critical-line zeros. Thus all finite counts above have the interpretations (1.1)–(1.3).

For completeness, the Riemann–von Mangoldt formula gives

$$(6.5) \quad N(X) = \frac{X}{2\pi} \log \frac{X}{2\pi} - \frac{X}{2\pi} + O(\log X) = \frac{X \log X}{2\pi} + O(X),$$

with the usual endpoint convention; see [19] or [4, Lemma 2]. The simple-real span obeys $S \leq X \log X / (2\pi)$, including when it is defined to be zero. Therefore

$$(6.6) \quad \limsup_{X \rightarrow \infty} S/N(X) \leq 1.$$

Only an upper bound is needed, since every span debit is nonnegative.

6.2. Smoothing without changing the test with height. The densities (5.1) have sharp endpoints and cannot simply be inserted as smooth strict-support windows. We now justify their use in the final constants. Fix $j \in \{0, 1\}$, a dilation $0 < \sigma < 1$ and a slack $0 < \delta < 1$. Let $F = f_{j,\sigma}$ and let ψ be a nonnegative even smooth cutoff supported strictly inside $(-\sigma/2, \sigma/2)$. Set

$$(6.7) \quad f' = \frac{\psi^2 F}{\int \psi^2 F}, \quad \eta' = \psi \sqrt{F / \int \psi^2 F}.$$

Choose cutoffs tending to one on every compact interior subinterval, bounded between zero and one. Strict positivity and smoothness of F on that interior make η' smooth after extension by zero. The densities f' tend to F in L^1 and L^2 , and their energies tend to $R_j(\sigma)$; the second part of (5.2) is continuous in L^1 on a common compact support.

If $\tau = \|f' - F\|_1$, their Fourier transforms differ by at most τ on the *real* axis. Both have absolute value at most one there. The sum of all coefficients in a seven-point row is 12, so the squared-kernel energy changes by at most 24τ . Choose the cutoff sufficiently close that $24\tau \leq \delta a_j$. Applying the original local row to the dilated gaps gives

$$(6.8) \quad V_{7,f'}(g) + \sigma \frac{d_j}{6} \sum_{i=1}^6 g_i \geq a'_j := (1 - \delta) a_j.$$

Only real transform arguments occur in a simple-real Gram block, so no uniform comparison at complex zeta arguments is being assumed.

The common-witness transfer now reads

$$(6.9) \quad D_t(M_{f'}) \geq a'_j n - \sigma d_j S - 6a'_j.$$

The pressure has become σd_j ; there is no division by $1 - \delta$. At $j = 0$ use $t = 1$. At $j = 1$ use $t = 2$ for multiplicity two, and for the union recompute

$$(6.10) \quad \Gamma' = 4 + \sqrt{8(1 - a'_1)}, \quad t' = \Gamma'/2 - 1, \quad Z' = \Gamma'^2/8.$$

Here $t' > 2$ and the cap remains valid. The coefficient of u in (2.15) is still exactly zero.

Every application of (6.1) and (6.4) has the smooth density, dilation, slack and threshold fixed before $X \rightarrow \infty$. One then removes the cutoff at fixed slack, sends $\delta \downarrow 0$, and finally sends $\sigma \uparrow 1$. These successive limits of numerical lower bounds require no uniform analytic error as the test changes. An arbitrarily small fixed upper-budget margin for the distinct ratios is handled in the same way below.

7. EVALUATION OF THE GLOBAL PROPORTIONS

7.1. The simple-critical and total-distinct counts. Use the smoothed cosine density in (2.16), with $a = a'_0$, $d = \sigma d_0$ and $B = 6a'_0$. Since $e \geq 0$, the height limit, (6.4) and (6.6) give

$$\liminf_{X \rightarrow \infty} \frac{n(X)}{N(X)} \geq \frac{2 - \mathcal{R}(f') - \sigma d_0}{1 - a'_0}.$$

Take the successive approximation limits described in Section 6.2; this proves (1.8) with its exact constant C .

To obtain the distinct count, substitute the *full* finite inequality into (2.13) before dropping e :

$$\begin{aligned} N^d &\geq \frac{1}{2} \left\{ N + \frac{2N - H - \sigma d_0 S - 6a'_0}{1 - a'_0} + \left(\frac{2}{1 - a'_0} - 1 \right) e \right\} \\ &\geq \frac{1}{2} \left\{ N + \frac{2N - H - \sigma d_0 S - 6a'_0}{1 - a'_0} \right\}. \end{aligned}$$

The coefficient discarded is positive. The same limits now prove (1.9).

7.2. The union and higher-multiplicity mass. Use the smoothed two-mode density and (6.10). Equation (2.15) gives

$$Z'e \leq H + \sigma d_1 S + 6a'_1 - (1 + a'_1)N.$$

For this fixed smooth test it follows that

$$\limsup_{X \rightarrow \infty} \frac{e}{N} \leq \frac{\mathcal{R}(f') + \sigma d_1 - 1 - a'_1}{Z'}.$$

Equation (2.11) and the successive approximation limits give (1.10). Equation (2.12) gives (1.13) for each fixed $r \geq 3$. Although (2.12) is also valid at $r = 2$, the direct estimate (4.5), at $t = 2$ with the same smoothed row, is stronger there and yields (1.11).

7.3. Distinct low-multiplicity locations. A fixed finite budget avoids any uncontrolled division of an $o(N)$ term by N^d . Fix $\xi > 0$ and, for a fixed smoothed two-mode density, set

$$\mathcal{T} = \mathcal{R}(f') + \sigma d_1 + \xi.$$

Equations (6.4) and (6.6), together with $6a'_1/N \rightarrow 0$, show that $H + \sigma d_1 S + 6a'_1 \leq \mathcal{T}N$ for all sufficiently large X . For sufficiently small ξ , slack, cutoff error and $1 - \sigma$, the strict versions of the guards (4.9) hold. Apply the finite ratio (4.10) first, then pass to the height limit and remove ξ and the approximation parameters. This proves (1.12).

For each fixed $r \geq 3$, use $c = \Gamma'/2$ and (4.11)–(4.12). The same procedure and (4.13) give (1.14). The limiting guards can be checked without delicate numerical signs. For example,

$$\frac{5}{4} \leq R_0 < \frac{4}{3}, \quad 0 < \Delta < \frac{109}{250000}, \quad 1 < T_1 < 1.35, \quad 6 < \Gamma < 7$$

give $\beta > 0$, $3 - T_1 - a_1 > 0$, $8 - 2T_1 - a_1 > 0$ and $\alpha = \Gamma - T_1 > 0$. Moreover, using $c^2 = 2Z$ and $\alpha = Z - \beta$,

$$4\alpha + \beta - c^2 = 2Z - 3\beta > 0,$$

since $Z > 5$ and $\beta < 0.35$. The remaining guard $q_c = 0$ is the exact identity (2.17). Thus the strict conditions persist under the small perturbations just used, with this identity retained symbolically. The fine decimal enclosures are independent of these coarse qualitative checks.

One can also see directly why the denominator cannot become too small for a fixed test. Cauchy–Schwarz on the nonzero eigenvalues of A and its rank-one decomposition by distinct locations give

$$(7.1) \quad N^2 \leq \text{rank}(A)H \leq N^d H.$$

Since $H = O_f(N)$, this implies $N^d \gg_f N$. We nevertheless used a fixed upper budget so that the per-location inequalities remain exact throughout the argument.

7.4. Exact arithmetic and strict comparison signs. The numerical constants in the introduction are enclosed by rational intervals, not by floating-point rounding to a presumed number of correct digits. The verifier constructs π from Machin’s identity and alternating arctangent series, bounds square roots by rational squaring, and evaluates sine and cosine with explicit Taylor remainders. Interval division is used only after excluding zero from the denominator. It evaluates (1.4)–(1.7) and the five exact expressions independently of the local cover program. In particular it checks the cap inequalities and the strict margins between each exact lower endpoint and its displayed decimal.

The signs of the improvements over the named basic formulas also have a short analytic proof. Put

$$Z_0 = 3 + 2\sqrt{2}, \quad \eta_* = a_1 - d_1 - \Delta, \quad B = (R_0 - 1)/Z_0.$$

Then $\beta = R_0 - 1 - \eta_*$. Since $a_1 < 1/2$,

$$0 < Z_0 - Z = a_1 \left(1 + \frac{2\sqrt{2}}{1 + \sqrt{1 - a_1}} \right) < 3a_1.$$

Also $\eta_* > 633/125000$, whereas

$$\frac{(R_0 - 1)(Z_0 - Z)}{Z_0} < \frac{a_1}{5} = \frac{7}{2000} < \frac{633}{125000}.$$

It follows that $E < B$, which proves the strict union improvement and the stated higher-cutoff family comparisons. The estimates $1 < R_0 < 4/3$ used here follow from $\cos z \geq 3/4$, $\sin z/z \leq 1$, and, for the upper bound, the alternating-series inequality $\cos z < (5/6)(\sin z/z)$ at $z^2 = 1/2$. For example $\cos z \leq 1 - z^2/2 + z^4/24 = 73/96$, while $(5/6)(\sin z/z) \geq (5/6)(1 - z^2/6) = 55/72 > 73/96$.

The comparison with the mass-two formula $(3 - R_0)/2$ has numerator

$$K_* = a_1(3 - R_0) - 2(\Delta + d_1) > \frac{7}{240} - 2 \left(\frac{109}{250000} + \frac{3}{250} \right) > 0$$

over the positive denominator $2(2 - a_1)$. The comparison with the distinct-location formula $1 - (R_0 - 1)/(8 - 2R_0)$ has numerator $3K_*$ over $(8 - 2R_0)(8 - 2T_1 - a_1) > 0$. These identities explain the gain without relying on a long decimal expansion. The exact scalar calculation supplies the finer margins in Section 1.2.

8. SHORT-HEIGHT INTERVALS WITH FIXED SUPPORT MARGIN

Fix $0 < \theta < 1$ and let $H_X = X^\theta$. In this section all counts are restricted to $X < \gamma \leq X + H_X$, with their definitions and denominators otherwise unchanged. We append a subscript I when needed. For every fixed $0 < \sigma < \theta$ and fixed smooth real even test g supported in $[-\sigma, \sigma]$, Wang’s

Theorem 2.2 [21] states

$$(8.1) \quad \sum_{X < \gamma, \gamma' \leq X + H_X} \hat{g}\left(\frac{i(\rho - \rho') \log X}{2\pi}\right) \frac{4}{4 - (\rho - \rho')^2} \\ = \frac{H_X \log X}{2\pi} \left(g(0) + \int |\alpha| g(\alpha) d\alpha \right) + O_g(H_X + X^\sigma \log^2 X).$$

This is a separate analytic input, not a subtraction of two full-height correlation asymptotics. Its normalized error tends to zero for every fixed strict support margin:

$$\frac{H_X + X^\sigma \log^2 X}{H_X \log X} = O((\log X)^{-1} + X^{\sigma-\theta} \log X) = o(1).$$

Apply (6.2) to the two fixed tests G, G'' as before. The difference form of (6.5) gives

$$N_I(X) = \frac{H_X \log X}{2\pi} + O(H_X + \log X), \quad S \leq \frac{H_X \log X}{2\pi}.$$

It follows that $H_f/N_I \rightarrow \mathcal{R}(f)$ and $\limsup S/N_I \leq 1$. The finite argument is unchanged.

Define

$$(8.2) \quad T_j(\theta) = \frac{A_j}{\theta} + (B_j + d_j)\theta \quad (j = 0, 1), \quad E_\theta = \frac{T_1(\theta) - 1 - a_1}{Z}.$$

Smooth first at a fixed $\sigma < \theta$, take the height limit with every test fixed, remove the cutoff and floor slack, and only then let $\sigma \uparrow \theta$. The resulting lower limiting bounds are

$$(8.3) \quad \liminf \frac{n_I}{N_I} \geq \frac{2 - T_0(\theta)}{1 - a_0}, \quad \liminf \frac{N_I^d}{N_I} \geq \frac{1}{2} \left(1 + \frac{2 - T_0(\theta)}{1 - a_0} \right),$$

$$(8.4) \quad \liminf \frac{U_I}{N_I} \geq 1 - 2E_\theta, \quad \liminf \frac{M_{\leq 2, I}}{N_I} \geq \frac{3 - T_1(\theta)}{2 - a_1},$$

$$(8.5) \quad \liminf \frac{M_{\leq r, I}}{N_I} \geq 1 - \frac{r+1}{r-1} E_\theta \quad (r \geq 3 \text{ fixed}).$$

All lower limits here are as $X \rightarrow \infty$. A negative right side may be replaced by zero. These mass and total-distinct statements hold for each fixed $0 < \theta < 1$ under the stated short-height input, although at small θ some give no nontrivial lower bound.

For distinct low-multiplicity locations, the parameter guards must also survive this substitution. The bound

$$(8.6) \quad \liminf \frac{D_{\leq 2, I}}{N_I^d} \geq 1 - \frac{T_1(\theta) - 1 - a_1}{8 - 2T_1(\theta) - a_1}$$

holds whenever the strict versions of the nonconstant guards in (4.9) hold with $a = a_1$ and $\mathcal{T} = T_1(\theta)$. More explicitly, require

$$T_1(\theta) - 1 - a_1 > 0, \quad 6 - T_1(\theta) > 0, \quad 3 - T_1(\theta) - a_1 > 0, \quad 8 - 2T_1(\theta) - a_1 > 0;$$

the fixed condition $1 - 2a_1 > 0$ already holds. For $r \geq 3$, the analogous formula is

$$(8.7) \quad \liminf \frac{D_{\leq r, I}}{N_I^d} \geq 1 - \frac{E_\theta}{r - 1 - rE_\theta},$$

provided (4.11)–(4.12) hold with $c = \Gamma/2$, $a = a_1$, $\mathcal{T} = T_1(\theta)$, strictly for the nonconstant inequalities and with $q_c = 0$ retained exactly. These conditions hold for θ sufficiently close to one by the global checks and continuity. We do not assert them for every $\theta \in (0, 1)$. The fixed upper-budget construction in Section 7 justifies the distinct denominator here as well. None of these statements is uniform as θ varies with height, and none sets $\sigma = \theta$ inside (8.1).

9. SCOPE OF THE ACCOMPANYING LOCAL EXTENSION

The finite transfer permits other window lengths, profiles and distance-capacity weights, but its premise remains a uniform local inequality. To make that boundary concrete, the supplement also studies a thirteen-point objective with six explicit Fourier modes, two gap pressures and target 31/5000. Scalar exclusion partitions the gap range of a hypothetical strict counterexample into short and long components. Complete local certificates establish the all-long region and every region with exactly one short gap, including all ten interior positions. Their proof uses disjoint triples and an exact convex quadratic lower bound on the remaining small cells.

The resulting lower bounds are 0.006424625 on the all-long region, 0.00623 on the two endpoint-one-short regions and 0.0062 on each interior-one-short region, on the precise domains specified in the supplement. Thus a remaining strict counterexample must have at least two short gaps. Whether such a configuration exists remains unresolved in this work. The full thirteen-point inequality is not asserted, and none of its prospective zero-count consequences is included in (1.8)–(1.14).

The distinction is structural. The common-witness inequality can transfer a proved local floor, and the bounded state correction can transfer a proved subaction. Neither constructs a missing floor or a missing potential. In the present application the two completed seven-point inputs already suffice for every global bound stated in the article. The larger-window result records an additional proved local reduction without enlarging those conclusions.

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NOTE ON AI USE

The author declares the use of consumer-grade Gemini 3.1 and GPT-6 models for prior-art search, developing some ideas and proofs, computer-assisted verification, drafting and LaTeX typesetting; the author takes full and sole responsibility for all contents of this work.

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— SUPPLEMENTARY INFORMATION —

**LOCAL CERTIFICATES FOR COMPENSATED GRAM BOUNDS:
THE TWO SEVEN-POINT INPUTS AND THE COMPLETED ONE-SHORT
THIRTEEN-POINT REGIONS**

R. ARUN CHANDRU

ABSTRACT. This supplement gives the mathematical specification and acceptance proofs for two computer-assisted inequalities on six nonnegative gaps. They are the cosine row with floor $19/5000$ and pressure $1/3000$, and the two-mode row with floor $7/400$ and pressure $1/500$. We explain interval arithmetic, removable singularities, whole-cell range minima, matrix lower bounds, and complete-domain accounting, with source-bound records of completed executions. A separate six-mode, thirteen-point problem is developed from its scalar exclusions and two analytic baselines. An adjacent-long-gap certificate and four completed triple bounds then exclude its all-long region and every one-short region, including both interior placements. The remaining multi-short region is not certified. No full thirteen-point inequality or additional global zero-count bound is asserted.

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1. SCOPE, DEPENDENCIES, AND THE MEANING OF A CERTIFICATE

There are three different Fourier kernels in this supplement. The first two supply the only local numerical inputs to the main manuscript. The third concerns a separate, unfinished thirteen-point problem. A floor, pressure allowance, or quadratic reserve proved for one kernel is not available for either of the others.

All the mathematics needed to understand the local certificates is given below. The accompanying verification package is a computational implementation of these arguments, not a substitute for them. Its manifest identifies the distributed sources and the provenance of recorded results. A digest establishes byte identity; by itself it does not establish an interval enclosure or a covering argument.

The trust boundary has three parts. First are the displayed analytic and finite-dimensional arguments. Second are a correctly implemented outward arithmetic backend and the source implementing the stated acceptance and coverage rules. Third is an actual completed execution of that source with the exact configuration in question. The recorded executions used Arb through `python-flint` version 0.9.0 at 128-bit precision for scalar cell tables; the later triple certificates use exact rational arithmetic after these tables have been established. None of these computations is presented as a formal proof-assistant verification.

The scalar-cell, range-minimum, and convex-tangent verification framework, including the cosine seven-point row, comes from Ainta's MIT-licensed implementation [1]. The two-mode callbacks and the additional acceptance rules are adaptations. The Fourier-profile and local-pressure viewpoint is not claimed here as an invention of this supplement. The main manuscript discusses its mathematical prior art and the distinct common-witness transfer. Here the purpose is to expose the precise local proof obligations and their evidence, including the limits of that evidence.

2. THE KERNELS AND THE LOCAL ENERGY

Put $z = 1/\sqrt{2}$ and define the normalized cosine density by

$$(2.1) \quad f_0(t) = \frac{\cos(\sqrt{2}t)}{\sqrt{2} \sin z} \quad (|t| \leq \tfrac{1}{2}), \quad f_0(t) = 0 \quad (|t| > \tfrac{1}{2}).$$

For a finite real vector $u = (u_1, \dots, u_d)$, set

$$(2.2) \quad f_u(t) = f_0(t) + \sum_{j=1}^d u_j \cos(2\pi j t) \quad (|t| \leq \tfrac{1}{2}),$$

and set it to zero elsewhere. Direct integration gives $\int f_0 = 1$; each perturbing cosine has integral zero. Thus $\int f_u = 1$. The three choices used here are

$$(2.3) \quad u_C = (), \quad u_H = \left(\frac{13}{500}, -\frac{7}{500} \right), \quad u_{13} = \frac{1}{4000}(37, -69, -1, 8, -11, 2).$$

The elementary inequalities $\cos z \geq 3/4$ and $0 < \sqrt{2} \sin z \leq 1$ imply $f_0 \geq 3/4$ on its support. The absolute coefficient sums in the last two profiles are respectively $1/25$ and $4/125$. Consequently all three densities are nonnegative; in particular $f_{u_{13}} \geq 359/500 > 0$. Each is real and even.

Our Fourier convention and squared kernel are

$$(2.4) \quad k_u(x) = \int_{-1/2}^{1/2} f_u(t) e^{-2\pi i x t} dt, \quad q_u(x) = k_u(x)^2 \quad (x \in \mathbb{R}).$$

The kernel is real on the real axis, and $q_u \geq 0$. Writing $\text{sinc } y = \sin y / y$ with $\text{sinc } 0 = 1$, direct integration yields the entire formulas

$$(2.5) \quad k_0(x) = \frac{\text{sinc}(\pi x - z) + \text{sinc}(\pi x + z)}{2\sqrt{2} \sin z},$$

$$(2.6) \quad k_u(x) = k_0(x) + \frac{1}{2} \sum_{j=1}^d u_j \{ \text{sinc}(\pi(x - j)) + \text{sinc}(\pi(x + j)) \}.$$

These formulas have no actual poles, even when a shifted argument vanishes. Since the support is compact, differentiation under the integral is valid to every order. Positivity and normalization give

$$(2.7) \quad |k_u| \leq 1, \quad |k'_u| \leq \pi, \quad |k''_u| \leq \pi^2, \quad |q''_u| \leq 4\pi^2.$$

The last inequality follows from $q''_u = 2(k'_u)^2 + 2k_u k''_u$.

For $m - 1$ nonnegative gaps $g = (g_1, \dots, g_{m-1})$, define

$$(2.8) \quad V_{m,u}(g) = \sum_{h=1}^{m-1} \frac{2}{m-h} \sum_{i=1}^{m-h} q_u(g_i + \dots + g_{i+h-1}).$$

Zero gaps are included. The coefficients in this formula are part of the objective and must not be replaced by uniform coefficients over all pairs. The two seven-point assertions are

$$(2.9) \quad V_{7,C}(g) + \frac{1}{3000} \sum_{j=1}^6 g_j \geq \frac{19}{5000},$$

$$(2.10) \quad V_{7,H}(g) + \frac{1}{500} \sum_{j=1}^6 g_j \geq \frac{7}{400},$$

for every $g \in [0, \infty)^6$. These are whole-orthant statements, not inequalities only at a mesh or at distinct configurations.

3. ARITHMETIC THAT ENCLOSES FUNCTIONS, NOT SAMPLES

3.1. Outward operations and removable singularities. An interval $[a, b]$ denotes a set containing the exact quantity in question. Sums, differences, products, and quotients are rounded outwards; division is allowed only after zero has been excluded from the denominator interval. For rational intervals the endpoints and all comparisons can be exact. For Arb balls the containment guarantee of the backend is used, followed, where required, by a downward conversion to the exact real number represented by a binary64 value. Converting a lower endpoint to nearest binary64 and then applying `nextafter` toward $-\infty$ gives a conservative lower endpoint when the result is finite. A nonnegative lower bound is clipped at zero only when nonnegativity has independently been proved.

Away from zero, the identities

$$\begin{aligned} \text{sinc}'(y) &= \frac{y \cos y - \sin y}{y^2}, \\ \text{sinc}''(y) &= \frac{(2 - y^2) \sin y - 2y \cos y}{y^3} \end{aligned}$$

may be evaluated by interval arithmetic. If an argument ball includes zero, these divisions are not permitted. The following explicit polynomials handle such balls. For $|y| \leq b < 1$, let

$$\begin{aligned} P_0(y) &= \sum_{n=0}^8 \frac{(-1)^n y^{2n}}{(2n+1)!}, \\ P_1(y) &= - \sum_{n=0}^8 \frac{(-1)^n y^{2n+1}}{(2n+1)!(2n+3)}, \\ P_2(y) &= - \sum_{n=0}^8 \frac{(-1)^n y^{2n}}{(2n)!(2n+3)}. \end{aligned}$$

Then the respective error bounds are

$$(3.1) \quad |\text{sinc} - P_0| \leq \frac{b^{18}}{19!}, \quad |\text{sinc}' - P_1| \leq \frac{b^{19}}{19! 21}, \quad |\text{sinc}'' - P_2| \leq \frac{b^{18}}{18! 21}.$$

For completeness, $\text{sinc } y = \int_0^1 \cos(ty) dt$. Differentiate this identity once or twice and apply the real Taylor remainder for sine or cosine inside the integral. Integrating the extra powers of t gives exactly the three denominators in (3.1). Thus these are bounds over the whole ball, not an alternating-series test at its center. Substitution in (2.5)–(2.6), with factors π and π^2 for differentiation in x , encloses k, k', k'' and hence q, q', q'' for each of the three specified profiles.

3.2. A completely rational point evaluator. The scalar gates and the atomic Taylor certificates below can be checked without floating transcendental arithmetic at their rational evaluation points. Here is an explicit construction. The Machin identity

$$(3.2) \quad \pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

and the sums $\sum_{n=0}^{40} (-1)^n / ((2n+1)a^{2n+1})$ give rational enclosures for the two arctangents. Each error is at most $1/(83a^{83})$, with its alternating sign. Combine their endpoint intervals outwards. If desired, after each operation round to the outward rational lattice of spacing 10^{-50} .

Put

$$(3.3) \quad R_0 = \frac{1}{2} + z \cot z.$$

The Taylor sums through index 30 for $\cos z$ and $\sin z/z$, with the next alternating term, enclose them using only powers of $z^2 = 1/2$. Divide these positive rational intervals to enclose $z \cot z$. No numerical square root is needed for this operation.

For rational x , write $x = n + v$, where $n = \lfloor x + 1/2 \rfloor$ and $-1/2 \leq v < 1/2$. Evaluate $\sin(\pi v)$ through degree 47 and $\cos(\pi v)$ through degree 46, including symmetric remainders $2^{49}/49!$ and $2^{48}/48!$. These follow from $|\pi v| < 2$ and the ordinary real Taylor theorem. Interval evaluation of these polynomials, and the factor $(-1)^n$, encloses $\sin(\pi x)$ and $\cos(\pi x)$.

The directly integrated point formulas are

$$(3.4) \quad k_0(x) = \frac{\cos(\pi x) - 2(R_0 - 1/2)\pi x \sin(\pi x)}{1 - 2\pi^2 x^2},$$

$$(3.5) \quad k_u(x) - k_0(x) = \sum_{j=1}^d u_j \frac{(-1)^j x \sin(\pi x)}{\pi(x^2 - j^2)}.$$

At $x = j$ the corresponding mode integral is exactly $1/2$ and is handled separately. Other denominators must be checked to exclude zero. All rational point evaluations below have $x \geq 5/6$. The atomic helper includes the separately guarded range $[9/10, 1)$. On that below-one range $1 - 2\pi^2 x^2 < 0$ and each $x^2 - j^2 < 0$; the interval implementation still verifies these signs before division. At the positive integers the removable resonances are handled explicitly. This is a point

evaluator with proved error bounds. A whole interval is enclosed either by the entire sinc formulas or by an additional derivative bound; point evaluation alone is never a domain certificate.

4. THE COMPLETE SEVEN-POINT VERIFICATION

4.1. Pressure compactification and scalar cell tables. Write $F_7 = V_7 + p \sum g_j$, with one of the exact pairs (ε, p) in (2.9)–(2.10). Since $V_7 \geq 0$, the region $\sum g_j \geq \varepsilon/p$ is settled by pressure alone. Every strict counterexample would therefore have all its coordinates less than ε/p . The common grid is $G = 4000$, with closed cells

$$(4.1) \quad J_i = [i/G, (i+1)/G] \quad (i = 0, 1, \dots).$$

Let $L = \lceil \varepsilon G/p \rceil$. The coordinate cells $0, \dots, L-1$ cover every possible strict counterexample; shared boundaries are included. The recorded tables have $L+8$ entries. They supply lower bounds

$$(4.2) \quad q(x) \geq Q_i \geq 0, \quad q''(x) \geq D_i \quad (x \in J_i)$$

where the second assertion is made only for finite, available entries. To obtain Q_i , enclose $k(J_i)$ by Arb, take a nonnegative lower bound for its absolute value, convert downwards, and square downwards. To obtain D_i , enclose $q''(J_i)$ and convert its lower endpoint downwards. An unavailable curvature prefix is represented by $-\infty$, not zero.

It is unnecessary for these tables to contain every possible summed distance in the six-dimensional box. If a sum reaches beyond the stored q table, the universally valid lower bound zero is used for that energy term. If it reaches beyond the stored curvature table, the corresponding derivative test is unavailable. No extrapolation or unproved zero-curvature assertion is made.

Before forming multidimensional roots, remove any coordinate cell for which

$$(4.3) \quad \frac{1}{3}Q_i + p\frac{i}{G} \geq \varepsilon.$$

This is valid because that coordinate's nearest-gap term in V_7 has coefficient $1/3$, and every discarded term of F_7 is nonnegative. Maximal consecutive surviving groups form scalar components. Their Cartesian products cover all remaining candidates.

4.2. Range indices and elementary lower bounds. Represent a box by six inclusive integer ranges $[a_j, b_j]$. Its physical coordinate interval is $[a_j/G, (b_j+1)/G]$. A sum of h consecutive coordinates is contained in the scalar cells from

$$(4.4) \quad a_i + \dots + a_{i+h-1} \quad \text{through} \quad b_i + \dots + b_{i+h-1} + h - 1,$$

inclusive. The final $h-1$ is essential. Omitting it would leave part of the closed sum interval unchecked.

Taking the minimum of every Q entry in this range gives a whole-box lower bound for its energy term. A sparse range-minimum structure can answer the same query using two overlapping power-of-two ranges whose union is the entire requested range. Overlap is harmless for a minimum; a gap would not be. Adding all the resulting lower bounds with the original positive coefficients and the pressure at lower coordinate endpoints gives a valid interval lower bound for F_7 . The target is rounded upwards and lower contributions downwards before any binary64 acceptance comparison.

4.3. Signed Hessian bounds and center tests. For each pair distance write $x_e = v_e^T g$, where v_e is the zero-one incidence vector of its consecutive gaps, and let $c_e = 2/(7-h_e)$, where $h_e = v_e^T v_e$. The exact Hessian is

$$(4.5) \quad \nabla^2 F_7(g) = \sum_e c_e q''(v_e^T g) v_e v_e^T.$$

If $\ell_e \leq c_e q''(v_e^T g)$ throughout a box, then the exact represented matrix

$$(4.6) \quad H_- := \sum_e \ell_e v_e v_e^T \preceq \nabla^2 F_7(g)$$

throughout it. Indeed, the difference is a sum of positive semidefinite rank-one matrices with nonnegative coefficients. Negative ℓ_e are allowed. When c_e is bracketed by binary64 endpoints, its lower endpoint must be used for a nonnegative curvature lower bound and its upper endpoint for a negative one; the product is then rounded downwards. This sign rule prevents an upward error in a negative Hessian contribution.

Let c be the rational center of the physical box, and let r_j be its rational coordinate radii. If rigorous LDL decomposition establishes H_- positive definite, F_7 is convex on the box. In particular,

$$(4.7) \quad F_7(g) \geq F_7(c) - \sum_{j=1}^6 |\partial_j F_7(c)| r_j.$$

The center value is bounded below and each derivative magnitude above by Arb. A floating LDL test may be used to decide whether to attempt the rigorous test, but it cannot accept a box. The rigorous LDL test uses the exact represented rational matrix and requires every pivot strictly positive, with all interval divisions justified.

The two-mode computation also uses the stronger lower quadratic. Taylor's integral remainder along the segment from c to g gives

$$(4.8) \quad F_7(g) \geq F_7(c) + \nabla F_7(c)^T (g - c) + \frac{1}{2} (g - c)^T H_- (g - c).$$

If $H_- = LDL^T$ is positive definite, minimizing the right-hand side over all of $\mathbb{R}^6$ gives

$$(4.9) \quad F_7(g) \geq F_7(c) - \frac{1}{2} \nabla F_7(c)^T H_-^{-1} \nabla F_7(c).$$

Solving $Ly = \nabla F_7(c)$ by interval forward substitution encloses the quadratic term as $\sum_j y_j^2 / D_j$. It is subtracted with an upper bound. The minimizing point need not belong to the box.

A further valid fallback discards positive curvature contributions. For $\ell_e < 0$, Cauchy-Schwarz gives $\ell_e v_e v_e^T \succeq \ell_e h_e I$. Thus $\mu I \preceq H_-$ with $\mu = \sum_{\ell_e < 0} h_e \ell_e \leq 0$. The lower Taylor bound becomes

$$(4.10) \quad F_7(g) \geq F_7(c) - \sum_j |\partial_j F_7(c)| r_j + \frac{\mu}{2} \sum_j r_j^2.$$

All quantities are enclosed outwards. Each of (4.7), (4.9), and (4.10) is an alternative lower bound. Their maximum is permissible; adding them would count the same cost more than once.

Center values may be cached because every center distance is exactly of the form $k/(2G)$ for an integer k . A cache key must encode this exact index, and the profile and precision must be immutable during the run. Caching complete enclosing balls does not change any acceptance rule.

4.4. Exhaustion of the domain. If a box passes none of the valid lower bounds, bisect its widest integer coordinate range. For $a < b$, the two children $[a, m]$ and $[m + 1, b]$, $m = \lfloor (a + b)/2 \rfloor$, partition its atomic cell indices and cover its closed physical interval. Continue until all roots have been discharged. A remaining atomic box that passes no test is unresolved, not accepted and not automatically a counterexample. Similarly, a node cap, exception, or interrupted execution is not a successful proof.

Uniform pressure and (2.8) are invariant under full reversal of the six gaps. A reversal reduction, if used, must keep one representative for each genuinely reversed pair. There is no arbitrary permutation symmetry. Source review and complete tree traversal establish coverage; identities such as

$$(4.11) \quad \text{roots} + \text{splits} = \text{accepted leaves}, \quad \text{visited nodes} = \text{splits} + \text{accepted leaves}$$

are useful independent consistency checks on a completed binary forest, but are not a replacement for that coverage argument.

5. COMPLETED VERIFICATION OF THE TWO SEVEN-POINT ROWS

The distributed verification package recomputed both seven-point inequalities with the exact configurations above. The following table records these completed executions. Both use grid 4000, 128-bit Arb arithmetic, and curvature start index 1001. Each successful execution exhausted its covering forest with no unresolved atomic cell.

Quantity	Cosine row	Two-mode row
Exact floor	19/5000	7/400
Pressure per gap	1/3000	1/500
Grid / Arb precision	4000/128	4000/128
Coordinate cutoff L	45600	35000
Scalar table entries	45608	35008
Curvature start index	1001	1001
Initial roots	378	1
Visited nodes	268856	129093
Splits	134239	64546
Accepted leaves	134617	64547
Pressure leaves	1597	602
Interval leaves	111698	38533
Derivative-test leaves	21322	25412
Maximum depth	32	59
Unresolved / pending	0/0	0/0

The cosine coordinate components, in inclusive cell indices, are

$$(5.1) \quad [3809, 4778], \quad [7221, 9363], \quad [10572, 44827].$$

There are $3^6 = 729$ Cartesian roots, of which $(729 + 27)/2 = 378$ represent the reversal orbits. The two-mode computation has the single component $[3443, 34516]$ and therefore one six-dimensional root. Its category named “tangent” in the saved report includes the valid derivative alternatives above; that label is not a claim that each such leaf used only the affine tangent formula. Its center-cache report has 1666485 hits and 2574 misses.

The scalar-table SHA-256 values, with binary64 table encoding fixed by the accompanying source and manifest, are as follows. Line breaking of a printed digest has no mathematical significance.

Cosine q : a9992300d2bf71665aa2b6bd2727e798624cd297103bb200c7f0ca2baea55a2c
 Cosine q'' : 766b97d9ba81dff915acaf1c1ba9938166c3191011e744424c12fd3a141bd52
 Two-mode q : f0be0477f6c4c98c7c3d879abf79f509fa637e9d4740fe8ee83bd4af52899580
 Two-mode q'' : aa35a51991d4a84c748e1908f7d95482ac507051d7c67b16c6919ee09f35377c

The cosine row has a completed ordinary execution of the distributed source. The two-mode row has completed ordinary and -00 executions: their entire receipts agree after removing only the optimization flag and the engine’s elapsed time. Source identities, table digests and machine-readable covering summaries accompany the package. These completed computations support (2.9) and (2.10) subject to the stated implementation and arithmetic trust boundary. An optimization-mode comparison is a consistency check, not an independent enclosure method.

6. A SEPARATE THIRTEEN-POINT OBJECTIVE

For the remainder of the supplement, $k = k_{u_{13}}$ and $q = k^2$ refer only to the six-mode profile in (2.3). Define

$$(6.1) \quad F(g) = V_{13, u_{13}}(g) + \sum_{j=1}^{12} p_j g_j, \quad g \in [0, \infty)^{12},$$

where

$$(6.2) \quad p_1 = p_{12} = p_e = \frac{811}{4800000}, \quad p_2 = \cdots = p_{11} = p_i = \frac{6359}{24000000}.$$

In particular,

$$(6.3) \quad P := \sum_j p_j = \frac{239}{80000}, \quad \varepsilon := \frac{31}{5000}.$$

We do *not* assert $F \geq \varepsilon$ on this entire orthant. The first task is to restrict where a strict counterexample could lie. All scalar restrictions in the next two sections are necessary conditions under the explicit hypothesis $F(g) < \varepsilon$.

Write $a = 1/(\sqrt{2} \sin z)$. Since $11/12 \leq \sin z/z \leq 1$, we have $1 \leq a \leq 12/11$. The convenient full-density bounds

$$(6.4) \quad m := \frac{71}{100} \leq f_{u_{13}}(t) \leq M := \frac{311}{275} \quad (|t| \leq \tfrac{1}{2})$$

follow from $\cos z \geq 3/4$ and $\sum |u_j| = 4/125 \leq 1/25$. The bounds apply to the perturbed density, not just its cosine part.

6.1. Monotonicity and the minimum-gap bootstrap. Differentiating the even Fourier integral gives

$$(6.5) \quad -k'(x) = 4\pi \int_0^{1/2} t f_{u_{13}}(t) \sin(2\pi xt) dt.$$

For $0 < x \leq 1$ this is strictly positive. For $1 \leq x \leq 27/25$, the sine is at least $1/2$ on $[1/8, 3/8]$: its argument belongs to $[\pi/4, 81\pi/100] \subset [\pi/6, 5\pi/6]$. This part of the integral in (6.5), before its factor 4π , is at least $71/3200$. A negative contribution can occur only for $t \geq 1/(2x) \geq 25/54$; its absolute value is at most $M(1/2)(1/2 - 25/54) = 311/14850$. Their difference is $1183/950400 > 0$. All other contributions are nonnegative. Hence k is strictly decreasing on $(0, 27/25]$.

The exact point evaluator in Section 3 verifies $k(5/6) > 0$ and $k(5/6)^2/6 > \varepsilon$. If some gap were at most $5/6$, monotonicity would make its nearest term alone exceed the target. Thus all gaps of a strict counterexample exceed $5/6$. More generally, suppose all such gaps already exceed d_0 , and $d_0 < d_1 \leq 1$ with $k(d_1) > 0$. A gap at most d_1 would imply

$$(6.6) \quad F(g) \geq \frac{k(d_1)^2}{6} + d_0 P.$$

The following outward lower endpoints exceed ε and prove the successive improved cuts. The pressure $d_0 P$ includes all twelve gaps exactly once; only one nonnegative nearest energy term is kept.

d_0	d_1	Lower endpoint in (6.6)
5/6	9/10	0.007009907197317624826482
9/10	91/100	0.006608495667780589234137
91/100	183/200	0.006356600267380480502615
183/200	917/1000	0.006262235820193320264541
917/1000	459/500	0.006214284540455943193710

Set $\tau_0 = 209/200$. Exact evaluation encloses

$$(6.7) \quad k(\tau_0) \in [0.015366680647566453432485, 0.015366680647566453432486]$$

and gives

$$(6.8) \quad \frac{k(\tau_0)^2}{6} - p_i(\tau_0 - 459/500) > 0.000005706104020715559434.$$

For $459/500 \leq g \leq \tau_0$, monotonicity, positivity, and (6.8) imply $q(g)/6 + pg \geq p\tau_0$ for either pressure p . For $g \geq \tau_0$ pressure alone proves the same assertion. Thus this first scalar baseline holds on the entire half-line $[459/500, \infty)$.

If all gaps exceed d_0 but one is at most $d_1 \leq \tau_0$, use the eleven other scalar baselines to obtain

$$(6.9) \quad F(g) \geq \frac{k(d_1)^2}{6} + \tau_0 P - p_i(\tau_0 - d_0).$$

The steps $(d_0, d_1) = (.918, .920), (.920, .922), (.922, .924)$ have respective outward lower endpoints

$$\begin{aligned} &0.006456625078131788200580, \\ &0.006352660139980702315412, \\ &0.006250599414089590385868. \end{aligned}$$

They exceed ε . Consequently every strict counterexample has

$$(6.10) \quad g_j > \delta := \frac{231}{250} \quad (1 \leq j \leq 12).$$

Each decimal just displayed is a terminating rational lower bound, not a claim about an unbounded numerical error.

6.2. Compact necessary domains. Put

$$(6.11) \quad B = \varepsilon - \tau_0 P = \frac{49249}{16000000}, \quad W_p(x) = \frac{q(x)}{6} + p(x - \tau_0).$$

The nearest cost $w_p(x) = q(x)/6 + px$ is at least both $p\tau_0$ and px . Since all higher-distance terms are nonnegative, a strict counterexample satisfies

$$(6.12) \quad \sum_j p_j(g_j - \tau_0)_+ < B, \quad W_{p_j}(g_j) < B.$$

The second inequality follows by using the other eleven baselines. In particular its coordinates and span satisfy

$$(6.13) \quad \begin{aligned} g_1, g_{12} &< \tau_0 + B/p_e = \frac{3124439}{162200}, \\ g_2, \dots, g_{11} &< \tau_0 + B/p_i = \frac{16103731}{1271800}, \\ \sum_j g_j &< 12\tau_0 + B/p_e = \frac{623616}{20275}. \end{aligned}$$

The final assertion uses $\sum g_j \leq 12\tau_0 + \sum(g_j - \tau_0)_+$ and $p_j \geq p_e$. It does not assume the gaps lie close to one or two.

7. THE EXACT ONE-DIMENSIONAL GATE

7.1. A point enclosure plus an analytic Lipschitz bound. For $x > 0$, integration by parts in the Fourier integral yields

$$(7.1) \quad k(x) = \frac{f(1/2) \sin(\pi x) - \int_0^{1/2} f'(t) \sin(2\pi x t) dt}{\pi x}.$$

Here $f = f_{u_{13}}$ on its support; the endpoint contribution is explicit, so no smooth extension by zero is assumed. The cosine part decreases on $[0, 1/2]$. Hence

$$\begin{aligned} |f(1/2)| + \int_0^{1/2} |f'(t)| dt &\leq a + \sum_j |u_j| + 2 \sum_j j |u_j| \\ &\leq C := \frac{27751}{22000}. \end{aligned}$$

For the perturbation this uses $\int_0^{1/2} |2\pi j \sin(2\pi jt)| dt = 2j$. It follows that $|k(x)| \leq C/(\pi x)$. Differentiating (7.1) gives

$$k'(x) = \frac{f(1/2) \cos(\pi x)}{x} - \frac{k(x)}{x} - \frac{2}{x} \int_0^{1/2} t f'(t) \cos(2\pi xt) dt,$$

so

$$(7.2) \quad |k'(x)| \leq \frac{C}{x} + \frac{C}{\pi x^2}.$$

The integral formula and $f \leq M$ also give $|k'(x)| \leq \pi M/2$.

For a closed rational interval $I = [l, r]$ with $l > 0$, let $c = (l + r)/2$, $h = (r - l)/2$, and let $[k_{\text{lo}}, k_{\text{hi}}]$ enclose $k(c)$ by Section 3. For an outward interval $[\pi_{\text{lo}}, \pi_{\text{hi}}]$, define

$$(7.3) \quad K = \min \left(1, \frac{C}{\pi_{\text{lo}} l} \right), \quad L = \min \left(\frac{\pi_{\text{hi}} M}{2}, \frac{C}{l} + \frac{C}{\pi_{\text{lo}} l^2} \right).$$

Then $k(I) \subset [v, w] = [k_{\text{lo}} - Lh, k_{\text{hi}} + Lh]$. Put

$$(7.4) \quad A_{\text{lo}} = \max(0, v, -w), \quad A_{\text{hi}} = \min(K, \max(|v|, |w|)).$$

These bound $|k|$ below and above on I . Therefore

$$(7.5) \quad \frac{A_{\text{lo}}^2}{6} + p(l - \tau_0) \leq W_p(x) \leq \frac{A_{\text{hi}}^2}{6} + p(r - \tau_0) \quad (x \in I).$$

Every operation here is on rational endpoints with outward rounding. This is a whole-interval argument derived from the derivative bound, not midpoint sampling.

7.2. Closed partition and retained components. For each of the two pressures, start with the closed interval $[\delta, \tau_0 + B/p]$. If the lower bound in (7.5) is at least B , exclude the whole interval. Equality is sufficient because the sublevel under consideration is strict. If its upper bound is less than B , retain the interval. Otherwise bisect it unless its width is at most $1/16384$; such unresolved boundary intervals are retained. Any emergency budget exhaustion must also retain pending intervals. Neither uncertainty nor termination may silently discard a possible counterexample.

The distributed gate checker completed both closed partitions. The endpoint gate has 50 excluded, 72 strictly passing, and 7 unresolved terminal intervals. The interior gate has 55, 74, and 10 respectively. No emergency budget was exhausted. Across both pressures, each run visited 534 cells and used 519 distinct exact midpoint values. A preliminary envelope-only test, using $0 \leq q \leq K^2$, can settle a cell without a point evaluation. Exact sorting, shared endpoints, and total interval lengths checked that the final leaves partitioned each initial domain. The ordinary and -00 executions agree in their mathematical output.

Merging retained adjacent leaves gives two closed necessary components for each pressure. Their exact endpoints are recorded here so that no floating approximation is used to define a domain:

$$(7.6) \quad \begin{aligned} S_e^{\text{gate}} &= \left[\frac{393403101917}{425197568000}, \frac{538442949829}{425197568000} \right], \\ L_e^{\text{gate}} &= \left[\frac{183665052303}{106299392000}, \frac{8164814247361}{425197568000} \right], \\ S_i^{\text{gate}} &= \left[\frac{770962919881}{833486848000}, \frac{1054606088081}{833486848000} \right], \\ L_i^{\text{gate}} &= \left[\frac{22558120767}{13023232000}, \frac{20694035057199}{1666973696000} \right]. \end{aligned}$$

The following slightly enlarged domains suffice for the local arguments:

$$(7.7) \quad S_e = [459/500, 317/250], \quad S_i = [459/500, 633/500], \quad L = [5/3, \infty).$$

The inclusions of (7.6) in the corresponding sets of (7.7) are exact rational comparisons. Thus a strict counterexample has an S or L label at every coordinate. This does not label every arbitrary nonnegative twelve-tuple: the strict-counterexample hypothesis is indispensable.

8. ANALYTIC SHORT AND LONG BASELINES, WITH RESERVES

8.1. **The short baseline** 53/50. We first prove, independently of any gate, that

$$(8.1) \quad \frac{q(x)}{6} + px \geq \tau_S p, \quad \tau_S = \frac{53}{50}, \quad x \geq 459/500,$$

for either pressure p . For $459/500 \leq x \leq 1$, the sine in (6.5) is nonnegative. Exact integration gives

$$\begin{aligned} -k'(x) &\geq m \frac{\sin(\pi x) - \pi x \cos(\pi x)}{\pi x^2} \\ &\geq m \frac{-\cos(\pi x)}{x} \geq m \cos(\pi/10) > \frac{19m}{20} > \frac{3}{5}. \end{aligned}$$

Here $x \geq .9$, $x \leq 1$, and $\cos(\pi/10) \geq 1 - \pi^2/200 > 19/20$ using $\pi^2 < 10$. For $1 \leq x \leq T := 1063/1000$, split the integral at $b = 1/(2x)$. On $[0, b]$ the positive part is at least $m/(4\pi x^2)$. On $[b, 1/2]$, use $|\sin(2\pi xt)| \leq 2\pi x(t - b)$ and $t \leq 1/2$ to bound the absolute negative part above by $M\pi(x - 1)^2/(8x)$. Multiplying by 4π gives

$$(8.2) \quad \begin{aligned} -k'(x) &\geq \frac{m}{x^2} - \frac{M\pi^2(x - 1)^2}{2x} \geq \frac{m}{T^2} - 5M(T - 1)^2 \\ &= \frac{3}{5} + \frac{366235595129}{62148295000000} > \frac{3}{5}. \end{aligned}$$

This proves $-k' > 3/5$ throughout $[459/500, 1063/1000]$.

The exact point evaluation

$$(8.3) \quad k(53/50) \in [0.001990173438104262028271, 0.001990173438104262028272]$$

has lower endpoint greater than both $a_S := 19/10000$ and $5p_i$. On $[459/500, \tau_S]$ we therefore have $k(x) \geq k(\tau_S) > 0$ and

$$\frac{d}{dx} \{q(x)/6 + px\} = k(x)k'(x)/3 + p \leq -k(\tau_S)/5 + p < 0.$$

The nearest cost is at least its value at τ_S , which is at least $p\tau_S$. For $x \geq \tau_S$, pressure alone proves (8.1). No derivative assertion beyond the stated finite interval is used.

There is also a useful explicit reserve. For $459/500 \leq s \leq \tau_S$ put $d = \tau_S - s$. Integrating (8.2) gives $k(s) \geq a_S + (3/5)d$. Consequently

$$(8.4) \quad r_{S,p}(s) := q(s)/6 + p(s - \tau_S) \geq \frac{a_S^2}{6} + (a_S/5 - p)d + \frac{3}{50}d^2.$$

All three coefficients are positive for both pressures.

8.2. The long baseline 203/100. Put $A = \sqrt{2} \cot z = 2z \cot z$. The same elementary sine and cosine estimates give $3/2 \leq A \leq 24/11$. Differentiating (3.4), with $v = \pi x$, yields

$$(8.5) \quad k'_0(x) = \frac{\pi}{(2v^2 - 1)^2} \left[(2Av^3 + (4 - A)v) \cos v - (2(A - 1)v^2 + 1 + A) \sin v \right].$$

On $5/3 \leq x \leq 41/20$ we have $v > 5$, $\cos v \geq 1/2$, $\sin v \leq \pi/20$, and $4 - A > 0$. The bracket is therefore at least $Q(v) = Av^3 - (\pi/20)(2(A - 1)v^2 + 1 + A)$. Using $\pi^2 < 10$, $A - 1 \leq 13/11$, $1 + A \leq 35/11$, and $v > 5$ gives

$$(8.6) \quad \frac{\pi Q(v)}{4v^4} \geq \frac{15}{82} - \frac{1}{2} \left(\frac{13}{550} + \frac{35}{27500} \right) = \frac{15}{82} - \frac{137}{11000} > 0.$$

For the first term we used $\pi A/(4v) = A/(4x) \geq 15/82$. Since $Q > 0$, replacing $(2v^2 - 1)^2$ by the larger $4v^4$ in the positive lower bound is legitimate. Hence (8.6) is a lower bound for k'_0 .

The perturbation has absolute coefficient sum $4/125$, and therefore

$$|(k - k_0)'(x)| \leq 2\pi \int_{-1/2}^{1/2} |t| \sum_j |u_j| dt = \frac{2\pi}{125} < \frac{44}{875}.$$

Combining this with (8.6) proves

$$(8.7) \quad k'(x) \geq \frac{15}{82} - \frac{137}{11000} - \frac{44}{875} = \frac{3}{25} + \frac{589}{3157000} > \frac{3}{25} \quad (5/3 \leq x \leq 41/20).$$

The exact point evaluation is

$$(8.8) \quad k(203/100) \in [-0.008907305837779524800319, -0.008907305837779524800318],$$

so $k(\tau_L) < -1/125$ for $\tau_L := 203/100$. For $5/3 \leq x \leq \tau_L$, put $d = \tau_L - x$ and integrate (8.7) to get $k(x) \leq -1/125 - (3/25)d$. Thus

$$(8.9) \quad r_{L,p}(x) := q(x)/6 + p(x - \tau_L) \geq \frac{1}{93750} + (1/3125 - p)d + \frac{3}{1250}d^2.$$

The linear coefficient is positive, since $1/3125 - p_i = 1321/24000000 > 0$. For $x \geq \tau_L$, nonnegativity of q and pressure suffice. We have proved the entire half-line baseline

$$(8.10) \quad q(x)/6 + px \geq \tau_L p \quad (x \geq 5/3)$$

for both pressures. Neither this proof nor the short proof uses derivative samples.

8.3. Pattern budgets and safe use of reserves. Fix any S/L label pattern permitted by the gates, and put $t_j = \tau_S$ on S coordinates and $t_j = \tau_L$ on L coordinates. Define

$$(8.11) \quad A_{\text{pat}} = \sum_j p_j t_j, \quad \Delta_{\text{pat}} = \varepsilon - A_{\text{pat}}.$$

For a strict counterexample, the two baselines and $w_{p_j}(g_j) \geq p_j g_j$ imply

$$(8.12) \quad \sum_j p_j (g_j - t_j)_+ < \Delta_{\text{pat}}, \quad g_j < t_j + \frac{\Delta_{\text{pat}}}{p_j}, \quad \sum_j g_j < \sum_j t_j + \frac{\Delta_{\text{pat}}}{p_e}.$$

If there are a long interior and b long endpoint coordinates, these constants have the closed form

$$(8.13) \quad \begin{aligned} A_{a,b} &= \frac{53}{50}P + \frac{97}{100}(ap_i + bp_e), \\ \Delta_{a,b} &= \frac{12133}{4000000} - \frac{97}{100}(ap_i + bp_e), \\ \sum_j t_j &= 12\frac{53}{50} + \frac{97}{100}(a+b). \end{aligned}$$

All 33 budgets, $0 \leq a \leq 10$, $0 \leq b \leq 2$, are positive: the largest baseline is the all-long baseline, whose deficit is $1083/8000000 > 0$. Thus these scalar budgets alone exclude no complete label pattern, and they do not make arrangements with the same counts equivalent for the longer-distance energy terms.

There is a useful optional closed span bound from the upper gate endpoints $U_{eS}, U_{eL}, U_{iS}, U_{iL}$ in (7.6). Let

$$\begin{aligned} Z_e &= (2-b)(U_{eS} - \tau_S) + b(U_{eL} - \tau_L), \\ Z_i &= (10-a)(U_{iS} - \tau_S) + a(U_{iL} - \tau_L), \\ z_e &= \min(Z_e, \Delta/p_e), \quad z_i = \min(Z_i, \max(\Delta - p_e z_e, 0)/p_i). \end{aligned}$$

Then $\sum g_j \leq \sum t_j + z_e + z_i$. Indeed, relax the positive excesses $v_j = (g_j - t_j)_+$ to $0 \leq v_j \leq U_j - t_j$, $\sum p_j v_j \leq \Delta$. Since $p_e < p_i$, exchanging interior excess for available endpoint excess improves the span at no greater cost. Spend the budget on the two endpoint coordinates first, then the interiors; this gives exactly $z_e + z_i$. This bound is stated non-strictly. It does not authorize an equality rejection appropriate only to the strict caps in (8.12).

The polynomial reserves (8.4) and (8.9) bound the nearest terms including their pressures. On a box they may replace a weaker lower bound for that same nearest term. They must not be added to an energy estimate that has already spent it. If all coordinates except j have nonnegative residual reserves σ_ℓ , a valid further necessary cap is $g_j < t_j + (\Delta - \sum_{\ell \neq j} \sigma_\ell)/p_j$. The reserve for coordinate j is deliberately absent from the sum.

9. ONE ADJACENT-LONG-GAP CREDIT

9.1. A pressure choice dominated by both actual choices. For $x \geq 5/3$ define

$$(9.1) \quad \phi_L(x) = \begin{cases} p_i(x - \tau_L), & x \leq \tau_L, \\ p_e(x - \tau_L), & x \geq \tau_L, \end{cases} \quad r_L^*(x) = q(x)/6 + \phi_L(x).$$

This is the minimum of the two actual long residuals. Hence it is nonnegative by (8.10) and no greater than either $r_{L,p}(x)$. The function ϕ_L is increasing and continuous, with its only slope change at τ_L .

The adjacent-long credit is the whole-domain inequality

$$(9.2) \quad r_L^*(x) + r_L^*(y) + \frac{2}{11}q(x+y) \geq K_L := \frac{3}{100000} \quad (x, y \geq 5/3).$$

Here is its analytic compactification and finite certificate. Let

$$(9.3) \quad P_L(d) = \frac{1}{93750} + \frac{1321}{24000000}d + \frac{3}{1250}d^2.$$

It increases for $d \geq 0$. If $x \leq 39/20$, then $d = \tau_L - x \geq 2/25$ and $r_L^*(x) \geq P_L(2/25) = 3043/10^8 > K_L$. If $x \geq \tau_L + K_L/p_e = 179033/81100$, its pressure residual alone pays K_L . The same applies to y . It remains to cover the closed square

$$(9.4) \quad [39/20, 179033/81100]^2.$$

9.2. The complete scalar tables and finite square. For the six-mode profile the distributed generator completed scalar tables using grid $G = 500$, 128-bit precision, and 15393 entries. The unused prefix below index 462 has nonnegative lower bound zero and unavailable curvature $-\infty$; entries from 462 onward were computed on the entire closed cells using Section 3. The complete-table digests are

q : 720514E4C09259C32629BBA740B5F7E3D61A3E85F6389AD7AE27BC61C0BB305C
 q'' : A67323D95F9407B91ACF10BB517C60D8F4B2BB9CB73DA8E4AFC2523630EA0CB2.

The package includes the generated tables and their source-bound generation record. Its format specifies the exact profile, closed-cell convention, entry count, precision, unavailable prefix and byte order. The generator rebuilds the enclosures; matching a stored digest alone does not establish them. No twelve-dimensional covering claim is made by this scalar calculation.

The square (9.4) is covered outwards by cell indices 975, ..., 1103, giving 129 choices for each coordinate. Let Q_i be its exact represented binary64 lower bound for q and put $R_i = \max(0, Q_i/6 + \phi_L(i/500))$. The maximum with zero is valid because the true residual is nonnegative. For cells (i, j) the test is the exact rational comparison

$$(9.5) \quad R_i + R_j + \frac{2}{11} \min(Q_{i+j}, Q_{i+j+1}) \geq K_L.$$

Both sum cells are present; they cover the full interval for $x + y$. The expression is symmetric, so all $129 \cdot 130/2 = 8385$ unordered pairs suffice. Each of the four triple invocations below recomputed all 8385 comparisons before using an LL replacement, both in ordinary and optimized Python. The smallest lower bound was

$$(9.6) \quad \frac{23653871141765599661283}{634106827533765836800000000} > \frac{3}{100000}.$$

Together with the closed tails, this proves (9.2) subject to the completed-table arithmetic boundary described above.

A use of (9.2) replaces its two nearest residuals and their connecting lag-two energy term by K_L . It does not add K_L while retaining those terms unchanged. In a longer string, such replacements can be made on disjoint adjacent pairs. For example, maximum matching credits w_j along a path can be selected by the elementary recurrence $D_0 = D_1 = 0$, $D_v = \max(D_{v-1}, D_{v-2} + w_{v-1})$. Two neighboring LL replacements sharing a middle residual may not be added. Only this disjoint replacement principle is needed below.

10. FOUR TRIPLE INEQUALITIES AND THEIR FINITE CORES

For labels $T_1, T_2, T_3 \in \{L, S\}$ define the local triple cost

$$(10.1) \quad C_{T_1 T_2 T_3}(x, y, z) = r_1(x) + r_2(y) + r_3(z) + \frac{2}{11} \{q(x+y) + q(y+z)\} + \frac{1}{5} q(x+y+z),$$

using r_L^* for every L and the actual pressure residual $r_{S,p}$ for each S. The four completed assertions are

Labels	Domains by label	Lower bound
LLL	$L = [5/3, \infty)$	$\lambda = 9/100000$
SLL_i	S_i, L, L	$\mu = 293723/2400000000$
LSL_i	L, S_i, L	$\mu = 293723/2400000000$
SLL_e	S_e, L, L	$\mu_e = 28447/480000000$

The short domains are those of (7.7). Full reversal gives the corresponding LLS assertions. Exchanging only the two long arguments of SLL is not a symmetry of (10.1); the two adjacent-sum slots would change.

Every residual and every remaining energy term in (10.1) is nonnegative on its domain. This permits a single variable's reserve to pay the whole target in a tail. For the first three rows, write η for their target. The long core is

$$(10.2) \quad [17/10, \tau_L + \eta/p_e].$$

Below its lower endpoint, $d \geq 33/100$ and $P_L(d) \geq P_L(33/100) > \max(\lambda, \mu)$ by exact rational comparison. Above its upper endpoint, $\phi_L(x) \geq \eta$. For an interior S, only the core

$$(10.3) \quad [47/50, 633/500]$$

remains: for $s \leq 47/50$, the positive polynomial in (8.4) at $d = 3/25$ already exceeds μ . The upper endpoint is the enlarged gate, not an extrapolated derivative range.

The inherited completed endpoint mixed cover uses the cores

$$(10.4) \quad S : [103/100, 317/250], \quad L : [189/100, 203/100 + 28447/81100].$$

The long lower tail is paid because $P_L(14/100) = 78495/1200000000 > \mu_e$. For the short lower tail, $d \geq 3/100$ and the last two terms of (8.4), with $p = p_e$, already exceed μ_e . The long upper endpoint is precisely $\tau_L + \mu_e/p_e$. All these cores and tails are closed; any overlap is harmless, and there is no omitted boundary between them.

11. EXACT ADAPTIVE AND ATOMIC TRIPLE VERIFICATION

11.1. An integer lower bound for whole boxes. The triple covers reuse the completed grid-500 tables. Round each finite rational core outwards to inclusive cell indices and take their Cartesian product. For an adaptive box $[a_j, b_j]$, use the full sum ranges in (4.4), now for one, two, or three coordinates. A range minimum supplies the lower q value over every required distance.

The interval and LL-replacement tests can be implemented with integer fixed-point arithmetic of scale $S = 2^{80}$. Set $T_i = \lfloor SQ_i \rfloor$, regarding the binary64 value Q_i as its exact rational value. For a single coordinate cell a safe residual integer is

$$(11.1) \quad \max(0, \lfloor T_i/6 \rfloor + \lfloor S\phi(i/500) \rfloor),$$

where ϕ is its actual short pressure residual or the piecewise long pressure in (9.1). Precompute this residual bound for every coordinate cell. For a wider coordinate interval, the implemented bound takes the minimum of those residual bounds over all its cells; each individual cell bound is valid on its entire closed cell. One could instead take a range minimum of T_i and use the leftmost pressure value, but that generally gives a weaker lower bound and is not the recorded residual-table rule. Flooring a negative pressure contribution means rounding toward $-\infty$, not truncating toward zero. Each of the two lag-two contributions and the lag-three contribution is also multiplied by its exact coefficient and floored downwards. Accept only if the sum of these integers is at least $\lceil S\eta \rceil$.

If two adjacent labels are L, one alternative bound replaces their two residuals and connecting lag-two term by $\lfloor SK_L \rfloor$, retaining the other residual and unused energy slots. When both adjacent pairs are LL, take the maximum of the two alternatives, not their sum. The ordinary interval lower bound is another alternative. Every accepted lower bound therefore uses each of the six slots of (10.1) at most once.

Bisect the widest nonsingleton integer coordinate range as in Section 4. Reversal can remove only boxes whose entire reflected box is represented by the retained half: for *LLL* and *LSL* the first and third label domains agree, whereas no internal reversal reduction is made in *SLL*. A valid implementation retains diagonal or overlapping ranges for further examination instead of discarding them on the basis of a center ordering.

11.2. A rigorous derivative from three exact point values. The preceding tests leave some atomic cells unresolved. The following exact quadratic test closes them without requiring a finer scalar grid. Let $h = 10^{-12}$. By (2.7), q' is Lipschitz with constant $4\pi^2$. The centered difference is its average over $[x - h, x + h]$, so

$$(11.2) \quad \left| q'(x) - \frac{q(x+h) - q(x-h)}{2h} \right| \leq \frac{4\pi^2}{2h} \int_{-h}^h |v| dv = 2\pi^2 h.$$

Using the rational point evaluator for $q(x-h)$ and $q(x+h)$, and $2\pi_{\text{hi}}^2 h$ as the error allowance, gives an exact rational interval for $q'(x)$. Cancellation may widen this interval, but cannot invalidate it. All required centers and their two probes are above .9. The guarded below-one formula in Section 3 is therefore available when a short midpoint is below one; an $x \geq 1$ routine is not silently applied outside its domain.

11.3. The full atomic Hessian and exact LDL test. The long pressure kink is at $\tau_L = 2.03$, exactly grid boundary 1015. Every closed atomic grid-500 cell lies on one side. Its appropriate affine branch agrees with the true pressure at every point of the closed cell, including a shared endpoint. The short pressure is already affine. Thus on an atomic triple cell, with coordinates $g = (x, y, z)$, the true cost has affine pressure and Hessian

$$(11.3) \quad \nabla^2 C(g) = \sum_{e=1}^6 c_e q''(a_e^T g) a_e a_e^T,$$

where the six pairs (a_e, c_e) are

$$(11.4) \quad \begin{aligned} &((1, 0, 0), 1/6), \quad ((0, 1, 0), 1/6), \quad ((0, 0, 1), 1/6), \\ &((1, 1, 0), 2/11), \quad ((0, 1, 1), 2/11), \quad ((1, 1, 1), 1/5). \end{aligned}$$

For an atomic tuple (i, j, k) , the lag-two curvature ranges include both indices $i+j, i+j+1$ and $j+k, j+k+1$. The lag-three range includes $i+j+k, i+j+k+1, i+j+k+2$. Let ℓ_e be the minimum of the corresponding complete curvature range, interpreted as an exact rational lower bound. Then

$$(11.5) \quad M = \sum_e c_e \ell_e a_e a_e^T \preceq \nabla^2 C(g)$$

throughout the atomic cell. Missing or nonfinite curvature disables this test; it is never changed to zero to obtain positive pivots.

Let $g_0 = ((2i+1), (2j+1), (2k+1))/(2G)$ be the rational center. Compute the LDL factorization $M = LDL^T$ with exact rational arithmetic. If any pivot is nonpositive, this criterion gives no acceptance. When every pivot is strictly positive, Taylor's integral remainder on the segment inside the box gives

$$(11.6) \quad \begin{aligned} C(g) &\geq C(g_0) + \nabla C(g_0)^T (g - g_0) + \frac{1}{2} (g - g_0)^T M (g - g_0) \\ &\geq C(g_0) - \frac{1}{2} \nabla C(g_0)^T M^{-1} \nabla C(g_0). \end{aligned}$$

The second line minimizes the lower quadratic over all of $\mathbb{R}^3$. Neither a stationary point of C nor the minimizing point of that quadratic needs to be inside the cell.

Enclose $C(g_0)$ and its gradient using the exact point values and (11.2). These evaluations use the actual unclipped residual functions in (10.1), not the maximum with zero used in an independent interval lower bound. For a gradient interval vector G , interval forward substitution gives $Y = L^{-1}G$. If $Y_j = [l_j, u_j]$, put

$$(11.7) \quad U = \sum_{j=1}^3 \frac{\max(|l_j|, |u_j|)^2}{D_j}.$$

All D_j are positive, so U is an upper bound for the quadratic debit despite dependencies between interval components. The exact rational acceptance condition is

$$(11.8) \quad C(g_0)_{\text{lo}} - U/2 \geq \eta.$$

No LL matching credit is added to this bound: the six original slots have already been used to define C . Failed pivots, inadequate lower bounds, domain-guard failures, or interrupted point evaluation produce no acceptance.

The adaptive cover is complete only when both its pending stack and its unresolved terminal list are empty. If a budget interrupt occurs during atomic work, the current box is restored to pending work. Node identities and exact covered atomic volumes are checked in addition to, not instead of, the source's exhaustive traversal.

12. COMPLETED TRIPLE EVIDENCE AND THE EXACT ASSEMBLY

12.1. What was actually executed. The completed endpoint mixed computation needed no atomic Taylor test. It visited 77 nodes with 38 splits and 39 accepted leaves: 37 interval leaves and 2 LL-replacement leaves. Ordinary and optimized executions completed the full outward core in (10.4).

The other three rows were completed with the atomic rule above, both in ordinary Python and with -00. The recorded counts agree in both modes:

Case	Nodes	Splits	Interval	LL	Taylor	Reversal
<i>LLL</i>	4117	2058	1059	9	868	123
<i>SLL_i</i>	2553	1276	1106	2	169	0
<i>LSL_i</i>	263	131	119	0	0	13

Every row has one initial root and zero pending or unresolved terminal boxes. The maximum depths are respectively 27, 27, and 22. The outward core atomic volumes are respectively 80621568, 45441792, and 45441792. They are volumes of adaptive leaves, not counts of point samples. In the symmetric rows, the proved and reversal-accounted volumes are respectively $78952385 + 1669183$ for *LLL* and $42186303 + 3255489$ for *LSL_i*; the *SLL_i* cover uses no reversal reduction.

These are completed executions of the distributed sources. All 1037 Taylor-accepted atomic cells were checked with rational matrix lower bounds, LDL factors, center and gradient intervals, and quadratic debits. The compact receipts retain exact canonical digests and counts of these generated objects, not the detailed matrices or point cache. The optional **-detailed** switch saves those objects in a fresh execution. The ordinary and optimized receipts agree after removal only of the optimization flag and the three declared timing fields. Checking the Taylor cells alone would still not prove an entire cover: the interval leaves, LL leaves, legitimate symmetries, roots, splits and zero pending volume remain indispensable.

12.2. The 78-term allocation. Partition the twelve gaps into the four consecutive triples

$$(12.1) \quad (g_1, g_2, g_3), \quad (g_4, g_5, g_6), \quad (g_7, g_8, g_9), \quad (g_{10}, g_{11}, g_{12}).$$

Let C_b^{actual} denote (10.1) with the actual pressure at every coordinate, rather than the dominated long choice r_L^* . With A_{pat} from (8.11), direct allocation of terms gives

$$(12.2) \quad F = A_{\text{pat}} + \sum_{b=1}^4 C_b^{\text{actual}} + R, \quad R \geq 0.$$

The allocation can be checked by the following inventory:

Distance in gaps	Original slots	Four triples	Remainder
$h = 1$	12	12	0
$h = 2$	11	8	3
$h = 3$	10	4	6
$4 \leq h \leq 12$	45	0	45
Total	78	24	54

Each used slot has its unchanged original coefficient, namely $1/6$, $2/11$, or $1/5$. Each pressure occurs once, through its baseline plus its residual. All 54 unused energy slots are nonnegative. By (9.1), each certified worst-pressure triple is no greater than C_b^{actual} .

If there is exactly one interior S, its baseline deficit is

$$(12.3) \quad \varepsilon - A_{\text{pat}} = \frac{1083}{8000000} + \frac{97}{100}p_i = \frac{941723}{2400000000} = 3\lambda + \mu.$$

The S is the middle coordinate of its block at positions 2, 5, 8, 11, using LSL_i . At positions 3, 4, 6, 7, 9, 10 it is an end coordinate, using SLL_i or its reversed LLS_i assertion. The other three blocks are LLL . Thus both interior placements are proved, and

$$(12.4) \quad F \geq \frac{31}{5000} \quad \text{on every retained interior-one-short region.}$$

No additional positive uniform margin is claimed here.

For either endpoint S, the mixed bound is μ_e . Its deficit is

$$(12.5) \quad \varepsilon - A_{\text{pat}} = \frac{143647}{480000000} = 3 \left(\frac{1}{12500} \right) + \mu_e.$$

Since $\lambda - 1/12500 = 1/100000$, the three all-long blocks now improve this to

$$(12.6) \quad F \geq 0.00623 \quad \text{on both retained endpoint-one-short regions.}$$

If all twelve gaps are L, all four triples use λ and

$$(12.7) \quad F \geq \tau_L P + 4\lambda = \frac{51397}{8000000} = 0.006424625.$$

There are thirteen labeled regions here: the all-long region and twelve one-short regions, seven classes under full reversal. This count is not a percentage of the full orthant that has been proved.

13. WHY THE MIXED CREDIT WAS REBALANCED, AND WHAT REMAINS OPEN

The stronger standalone interior mixed credit $\mu_{\text{old}} = 365723/2400000000$ is false. This can be demonstrated without an optimizer or a cover: evaluate the exact rational triple

$$(13.1) \quad (x, y, z) = (1.05423, 2.009397, 2.026457).$$

The point evaluator of Section 3 and the exact cost (10.1) give

$$(13.2) \quad C_{SLL_i}(x, y, z) \in [0.000132480370716898210350, 0.000132480370716898210351].$$

The upper endpoint is below μ_{old} and above the new target μ . Both long coordinates are below τ_L , so their worst-pressure residuals equal the actual interior-pressure residuals. The failure is therefore not an artifact of incompatible pressure choices. It refutes that local credit only, not the full objective (6.1).

The successful correction has a precise budget interpretation. Increasing each of the three LLL credits in a one-short decomposition from .00008 to .00009 lowers the required mixed credit by .00003. That is exactly $\mu_{\text{old}} - \mu$. The smaller mixed credit is established on its whole domain by the completed computations, not inferred from the single point (13.1).

Combining the scalar gate with (12.4)–(12.7) now gives the precise remaining boundary:

$$(13.3) \quad F(g) < 31/5000 \implies \text{at least two coordinates have a retained short label.}$$

Whether such a strict counterexample exists is unresolved here. The multi-short regions have not been certified. No universal thirteen-point floor, no extra global zero-count decimal, and no worldwide record-priority assertion follows from this local completion. In particular, the two seven-point inputs to the main manuscript remain exactly (2.9) and (2.10).

14. REPRODUCTION AND THE SCOPE OF THE PACKAGE

A self-contained reproduction has two logically distinct checks. The first evaluates the main manuscript's scalar count expressions from the two exact row configurations. Such an evaluation is conditional on the local rows; it is not a replay of them. The second runs a complete local verifier with the actual profile, pressure, floor, whole-cell tables, acceptance rules, and exhaustive coverage. The distributed package identifies these roles separately as `verify_counts.py` and `verify_local_rows.py`. Its README and manifest specify the commands actually supplied, their optional supplementary stages, and the execution status of each distributed source.

For the thirteen-point local results, the mathematical reproduction sequence is: the rational scalar bootstrap and closed gate; the short and long point checks together with the analytic derivative proofs; generation or full revalidation of the six-mode scalar tables; the LL square; the endpoint mixed cover; the three final triple covers with their atomic rule; and the exact assembly. The distributed `build_supplement_tables.py` constructs only the scalar tables. The entry point `verify_supplement.py` separates the cases `baselines`, `LL`, `LLL`, `SLL_i`, `LSL_i`, `SLL_e`, and `witness`; none of these is a full twelve-gap cover. The scalar-gate command and the completed receipt filenames are specified in the accompanying README. The program `assemble_supplement.py` binds the gate and all four source-generated triple receipts to the distributed sources and scalar data, then checks the exact regional allocation. It is a consistency-and-assembly checker, not an independent validator of arbitrary supplied proof logs. The reconstruction commands establish its local premises. Every stage is described above. A finite table can be distributed as an acceleration, but a verifier must know how to establish each of its whole-cell enclosures. Simply matching a stored digest establishes only that the same table has been read.

All acceptance guards must remain active under optimized Python; they must use explicit checked conditions rather than removable assertions. A bounded run may finish with an honest incomplete status. A modified program needs its own source-bound completed receipt before its output is described as a fresh replay.

This distinction is especially important here. The available completed records establish the two adopted seven-point rows and the specific thirteen-point local regions above, subject to the explicit computational trust boundary. They do not establish a complete thirteen-point target. The supplement supplies a linear, independently readable proof of what those records mean, while leaving that final unresolved question visible.

REFERENCES

- [1] Ainta, *zeta-simple-zeros*, MIT-licensed source code, commit 040c5e899e658aed7b56a2a87f501798fe10761d, <https://github.com/ainta/zeta-simple-zeros/tree/040c5e899e658aed7b56a2a87f501798fe10761d>. The original scalar-cell and convex-tangent engine and the cosine seven-point verification are attributed to this source; adapted portions retain its license.